OPERATOR MANUAL

Guide to the use of the interface, menus and operational functions

Product	Praetorian C2 Vehicle
Software version	1.0
Document edition	1.1 — 03 August 2026
Classification	UNCLASSIFIED

Support: praetorian.c2.it@gmail.com | moicanox.github.io/Praetorian-C2-Vehicle/

Document information

This manual describes the operational use of Praetorian C2 Vehicle through a logical workflow: system startup, Common Operational Picture interpretation, entity management, menu configuration, communications, tactical networks, video, mapping tools, logistics and protected data deletion.

Item	Purpose	Operating guidance
1.0	03/08/2026	Initial issue of the operator manual.
1.1	03/08/2026	Complete update with the new screens from the image file: main menu, top bar, Settings, Maps, Contacts, Video, Tools, entity options and Network subpages.

SCOPE This manual covers the functions represented in the available screens. Some items appear only when compatible sensors, radio links, TAK Servers, map datasets or mission files are configured.

OPERATOR RESPONSIBILITY Before use, verify authorizations, identity, network data, recipients and message content. Panic Zeroize and Selective Zeroize delete local data and must be used exclusively in accordance with organizational procedures.

Table of contents

1. Introduction and concept of operations

Praetorian C2 Vehicle is a command-and-control environment designed for vehicle-based operations. The map is the center of the interface: tracks, local objects, messages, sensors and planning tools are accessed through panels overlaid on the Common Operational Picture (COP).

1.1 System purpose

The system combines map display, Cursor on Target exchange, messaging, Data Packages, radio integration, video stream management, measurement tools and security functions. The operator must maintain constant awareness of network status, data origin and the time validity of displayed entities.

1.2 Operating principles

- Verify local identity, position, time and connection status before starting operations.
- Distinguish locally created data from data received from peers, servers, radios or external sensors.
- Check the recipient before sending chats, structured reports, packages or video.
- Adapt the frequency, content and size of data to the available bandwidth, especially on narrowband profiles.
- Do not change multiple parameters at the same time when diagnosing a network problem.
- Delete data only under an authorized order and procedure.

2. Startup, main menu and map interface

This section describes the initial operational screen, the main menu, the top bar and the basic map controls.

2.1 Main map screen

After startup, the COP is displayed full screen. Panels opened from the menus overlay the map without interrupting the operational context.

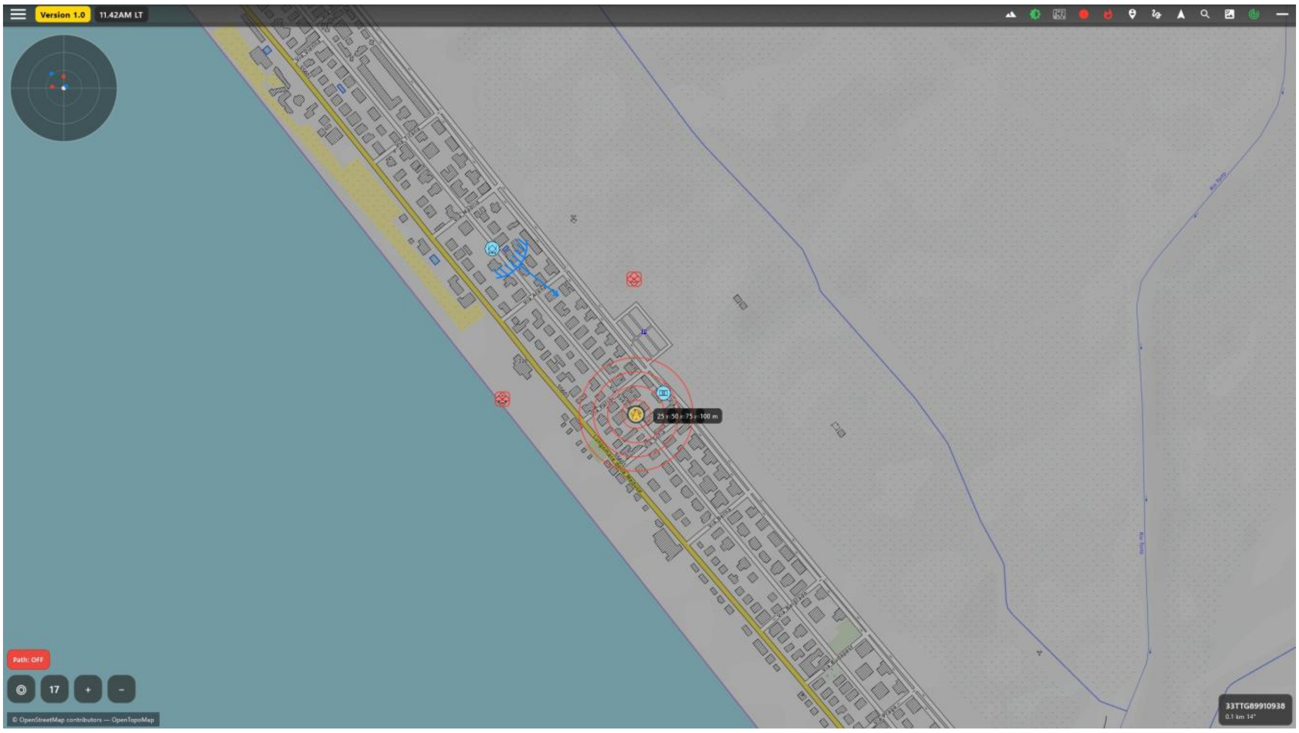

Figure 1 — Main interface with map, vehicle position, symbols and map controls.

Operating procedure

1. Verify that the map of the area is displayed and that the vehicle position is consistent.
2. Check the local time, version and status indicators at the top.
3. Use zoom and recentering before selecting a track or creating an object.

OPERATIONAL NOTE COP accuracy depends on the position source, the active map and the entity update frequency.

2.2 Main menu

The main menu organizes functions by activity: Alert, Settings, Network, Tools, Drawing Tools, FORM MSG, Chat, Video, Contacts, RADIO SA, Package, Camera, Maps, Logistic, Zeroize, About, Donate and Quit.

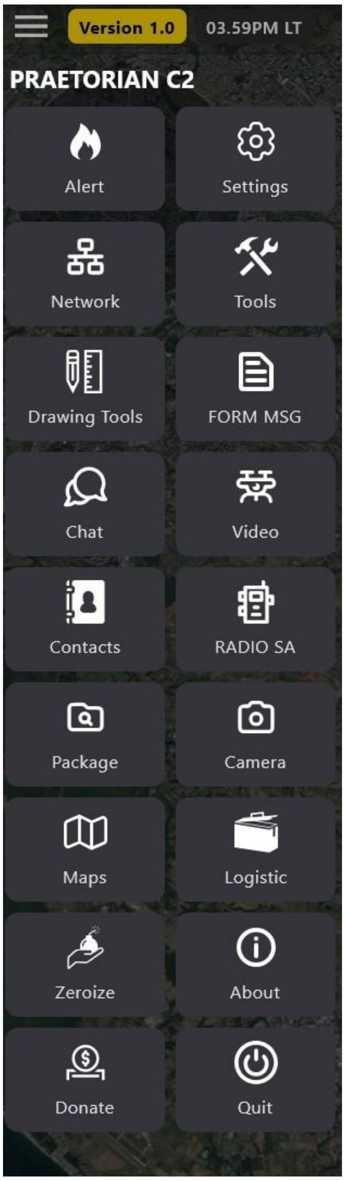

Figure 2 — Main menu providing access to the operational modules.

Item	Purpose	Operating guidance
Alert	Opens the preconfigured rapid messages.	Verify the preset text before use.
Settings	Configures identity, GPS, radar, presence, DTED, encryption, keyboard, photos, themes and logs.	Changes affect the behavior of the local client.
Network	Manages interfaces, inputs, outputs, servers and LEGIO v2.	Primary connectivity control.
Tools / Drawing Tools	Opens advanced tools and measurement/drawing functions.	Use the category appropriate to the task.
Chat / Contacts / Package	Manages conversations, peers and files.	Always confirm the recipient.
Zeroize	Deletes local data completely or selectively.	Irreversible action.

2.3 Top bar

The top bar contains indicators and contextual shortcuts. The visible icons depend on the active modules and system status.

Figure 3 — Top bar with shortcuts and status indicators.

Operating procedure

4. Check the colored indicators before starting operations.
5. Use the interface tooltips to identify unfamiliar icons.
6. Do not interpret a lit icon as confirmation of data quality: also check the relevant panel.

2.4 Map controls

The compact controls allow the view to be recentered, the zoom level to be read and path display to be enabled or disabled. Map sources are shown at the bottom.

Figure 4 — Recenter, zoom, path and map attribution controls.

Operating procedure

7. Use recentering to return quickly to the vehicle position.
8. Set a zoom level appropriate to the required level of detail.
9. Disable Path when the historical track is not needed or creates visual clutter.

3. Identity, position, tracks and symbology

Local and remote entities can be selected on the map to read attributes, start contextual actions and set motion prediction functions.

3.1 Selecting a track

Selecting a track highlights the spatial relationship between the vehicle and the entity. The information panel provides the identifier, distance, direction and available data.

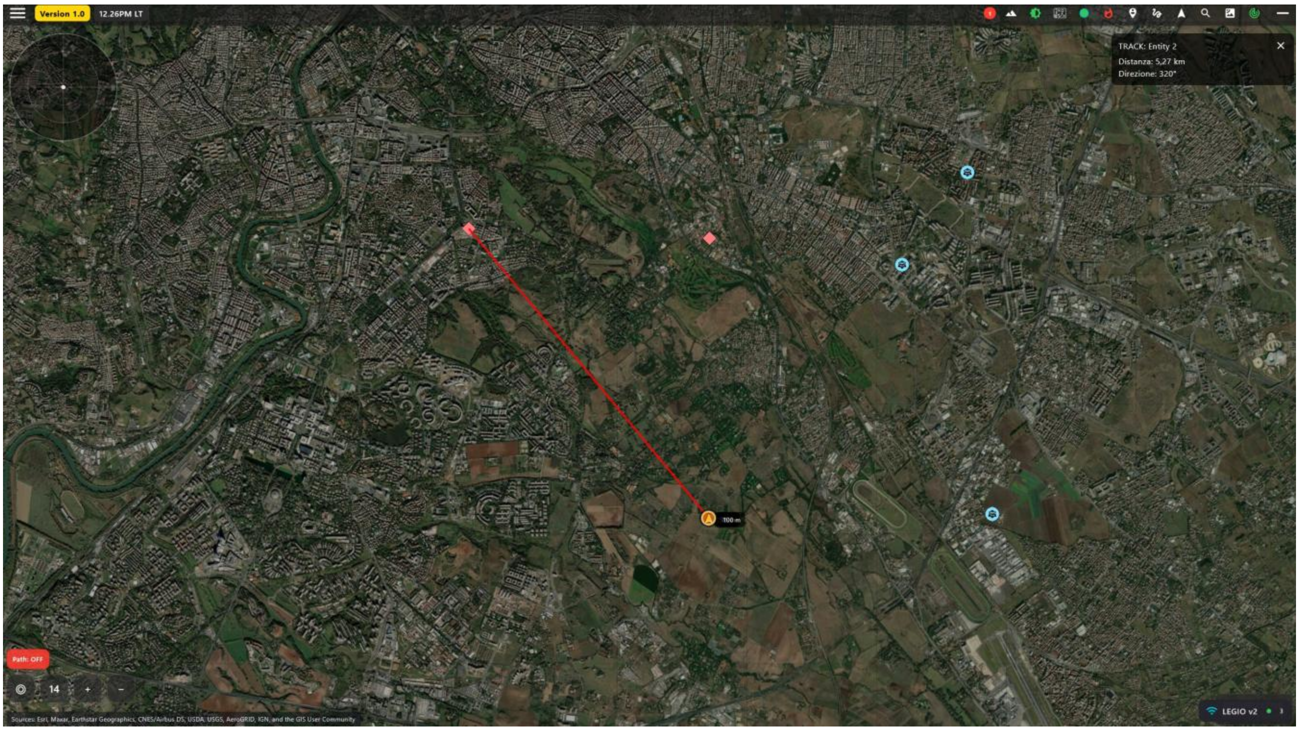

Figure 5 — Selected track with graphical link, distance and direction.

Operating procedure

10. Select the symbol on the COP.
11. Verify the callsign, affiliation, position and time of the last update.
12. Close the panel to remove the highlight.

CAUTION A track that has not been updated must be handled in accordance with organizational procedures; the displayed position may not represent the current situation.

3.2 Entity options panel

The contextual panel shows origin, MGRS and geographic coordinates, last update and icons for the available actions. Functions may vary according to the entity type.

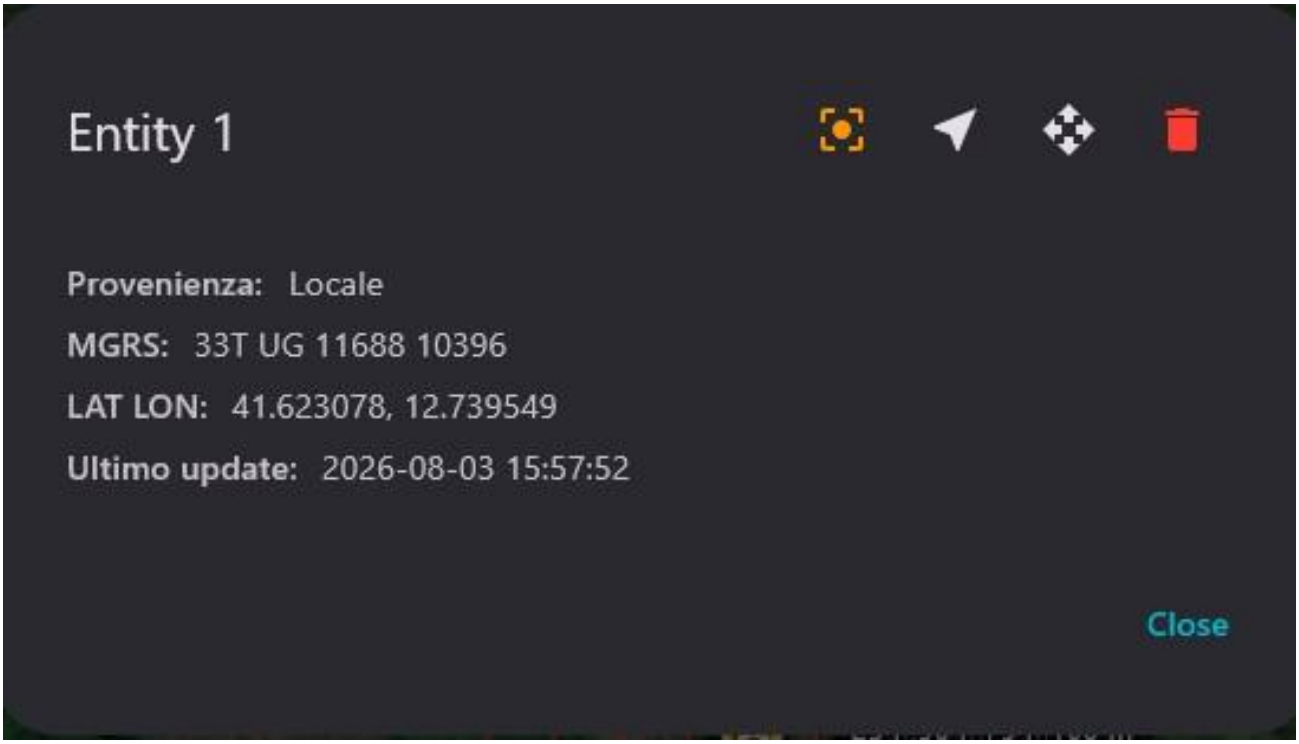

Figure 6 — Contextual panel for a local entity with position and quick commands.

Operating procedure

13. Check the Origin field to distinguish a local object from a received object.
14. Use coordinates only after verifying the format and the last update.
15. Delete a local entity only when it is no longer required and deletion is authorized.

3.3 Local dead reckoning

The Course & speed function allows a local entity to be moved in simulation by entering course and speed. Altitude and distance are optional when supported by the object type.

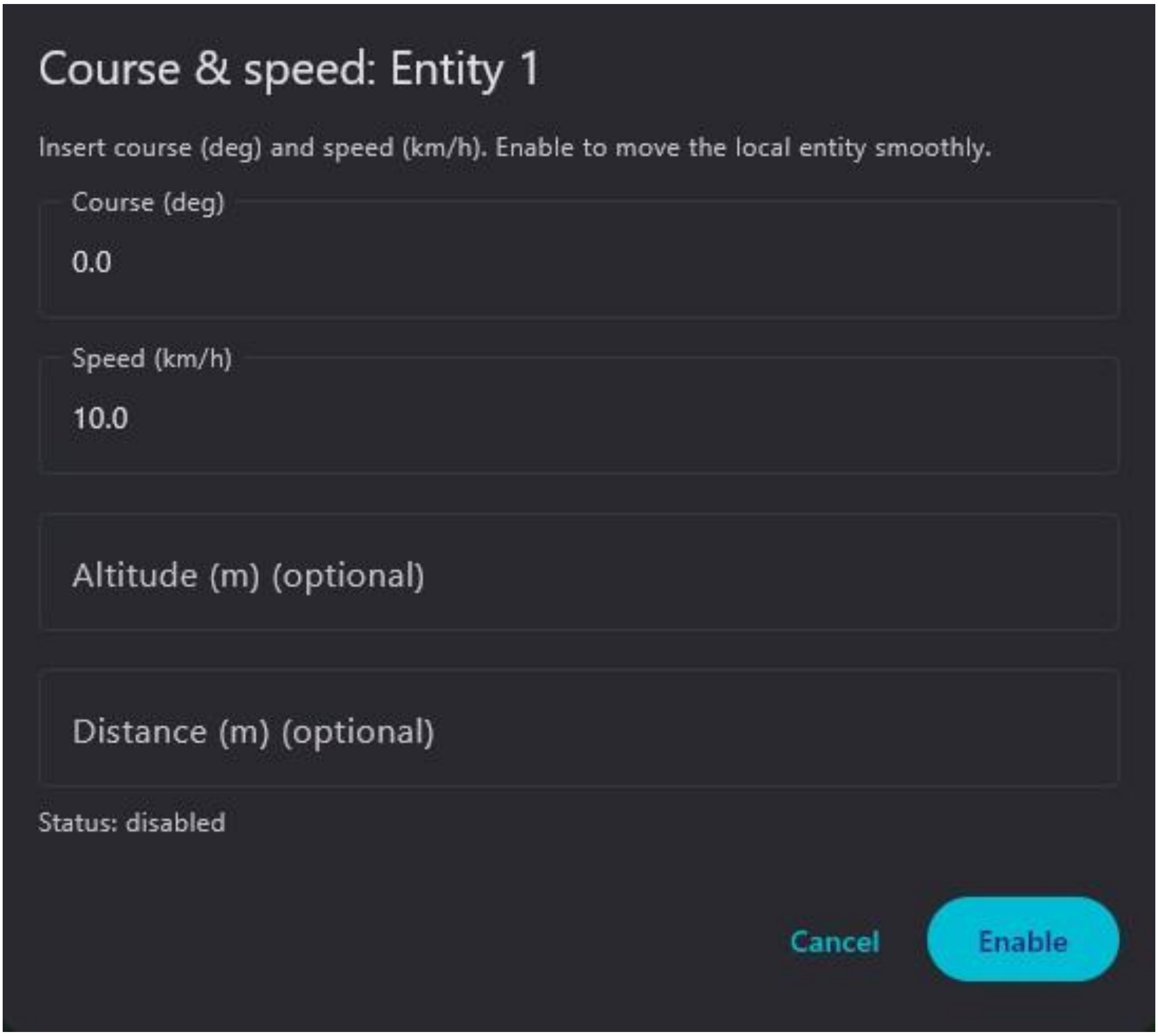

Figure 7 — Entry of course, speed, altitude and distance for motion prediction.

Operating procedure

16. Enter the course in degrees and the speed in the indicated unit.
17. Enter altitude or distance only when required by the scenario.
18. Press Enable and verify on the COP that the behavior is consistent.
19. Disable the function at the end of the simulation.

CAUTION The function does not replace a real tracking source and must not be interpreted as an observed update.

3.4 Entity symbol catalog

The selector allows the symbol to be filtered by affiliation and category, and then the item to be represented on the map to be selected.

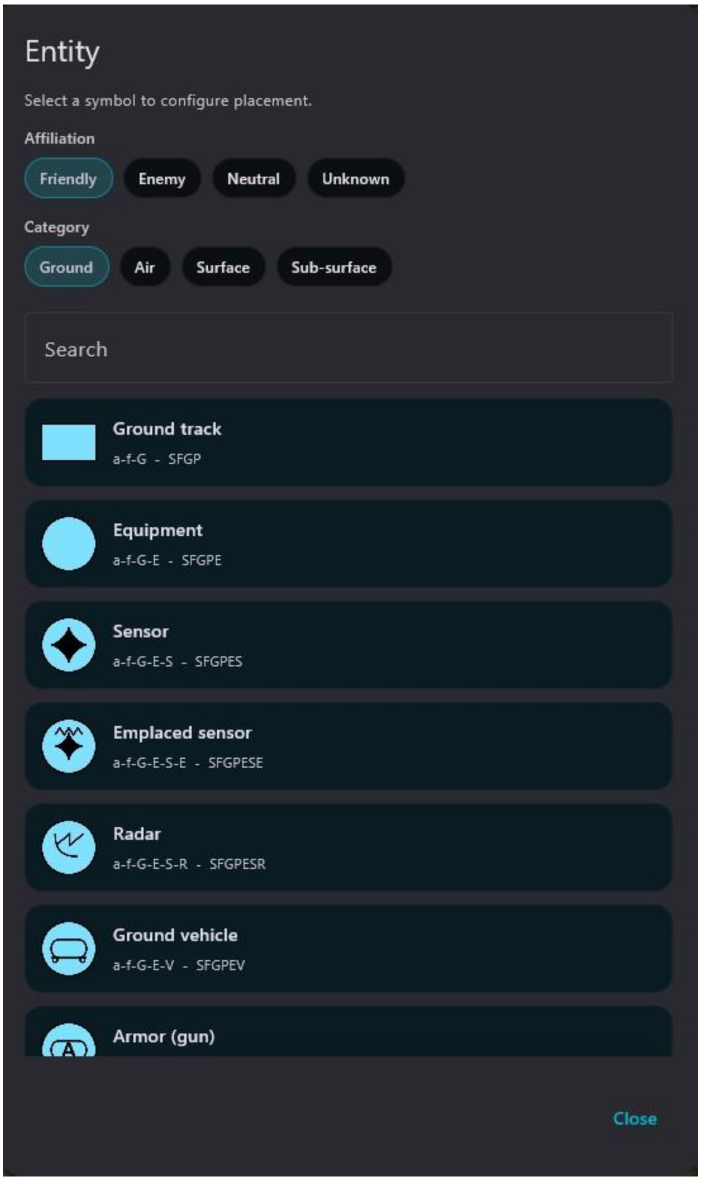

Figure 8 — Catalog for selecting the affiliation, category and symbol of an entity.

Operating procedure

20. Select the correct affiliation.
21. Choose the category and use search when available.
22. Confirm the symbol and verify its appearance on the COP.

CAUTION An incorrect affiliation or category may produce an incorrect operational interpretation.

3.5 My position panel

The My position panel shows the vehicle position and the time of the last update. Range fan and Weapon ranges can also be opened from the same panel.

Figure 9 — Summary of the local position in MGRS format and geographic coordinates.

Operating procedure

23. Compare MGRS and LAT/LON with the intended navigation source.
24. Check the time of the last update.
25. Open Range fan or Weapon ranges only after confirming the local position.

4. Settings and operator configuration

Settings centralizes the client's persistent configurations. Changes must be applied in a controlled manner and verified on the COP or in the relevant modules.

4.1 Settings overview

The screen shows the Self, GPS, Area of Interest Radar, Presence, DTED, Terrain Data, Crypto, Keyboard, Photo, Themes, Files and Logs cards. The summary status makes it possible to identify active or unconfigured modules quickly.

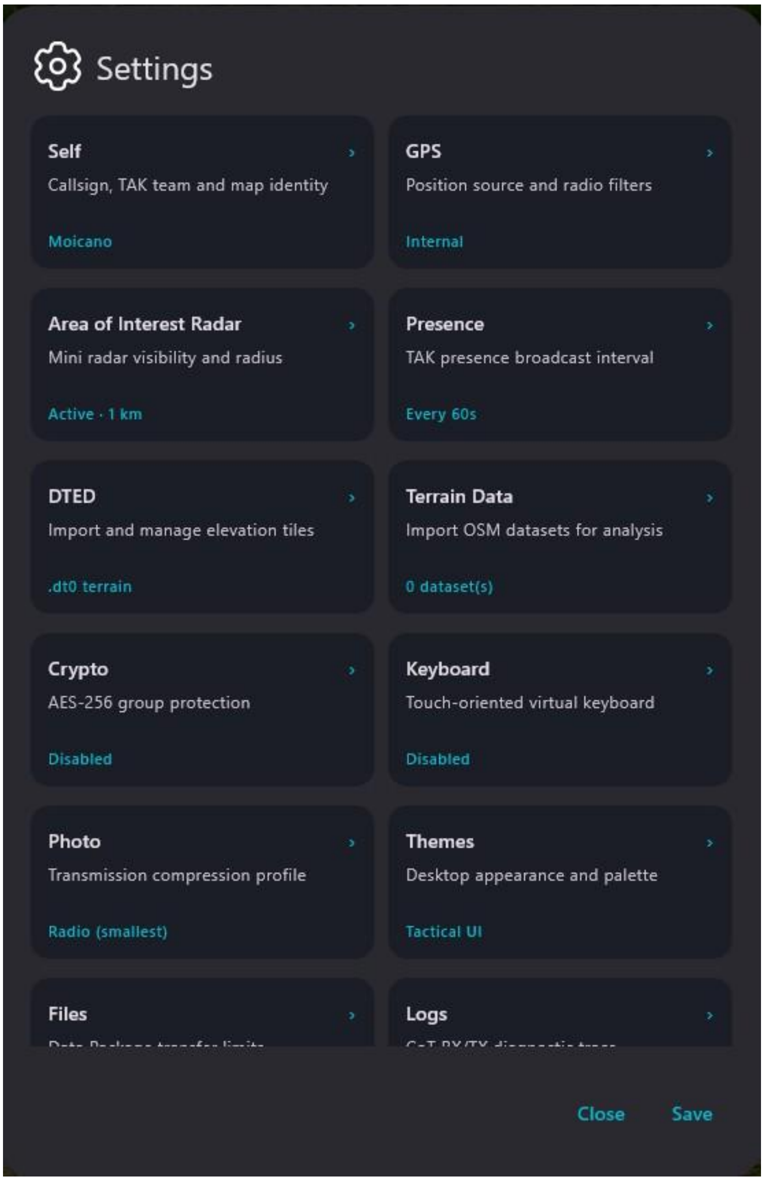

Figure 10 — Settings dashboard with summary status of the main configurations.

Operating procedure

26. Open the relevant card without changing other settings.
27. Apply Save only after verifying the values.
28. Return to the dashboard and check that the summary status has been updated.

4.2 Self: operator identity

Self defines the identity used to represent the vehicle and announce it to other systems. The screen allows the callsign, team color and local point color to be set.

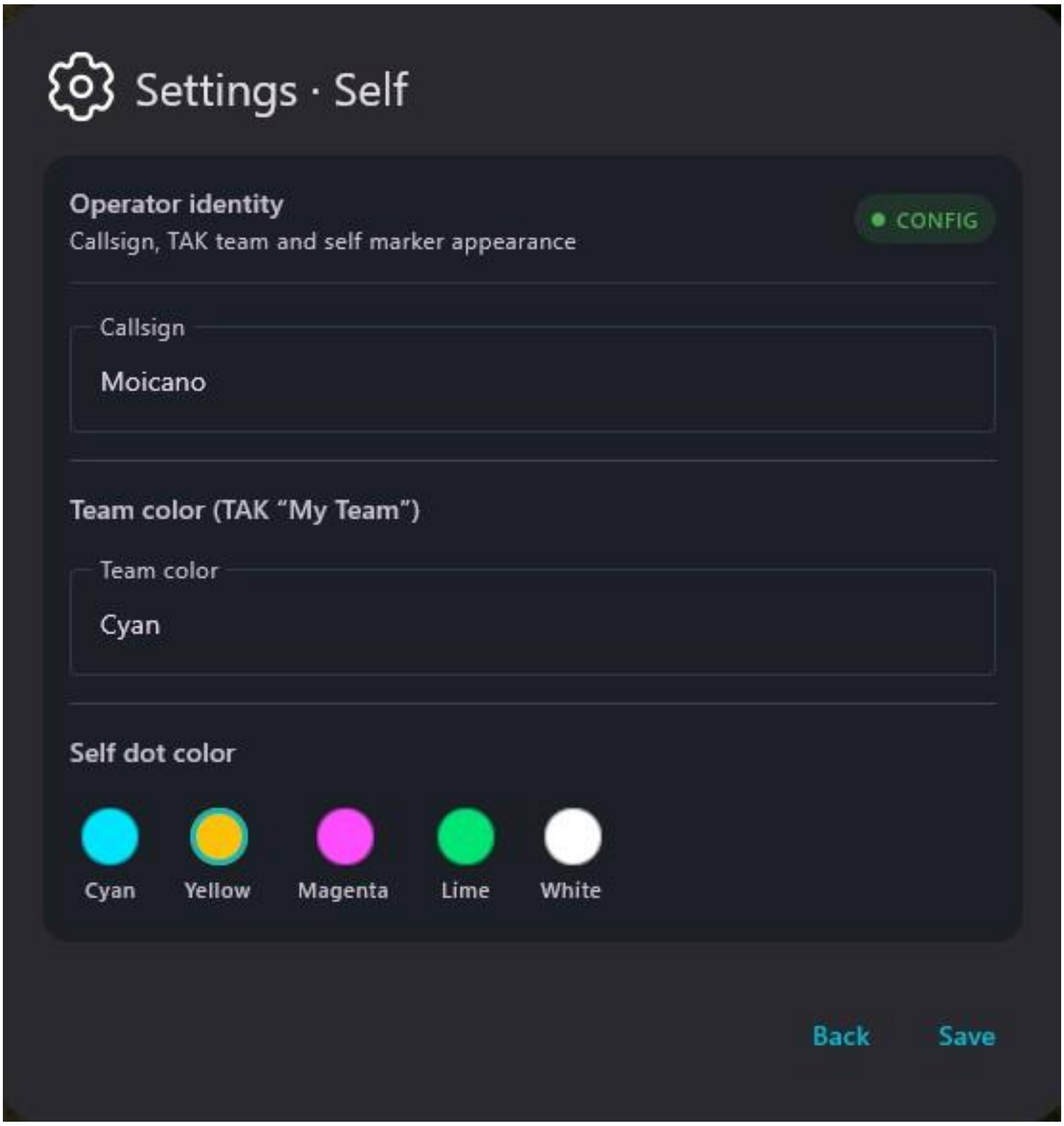

Figure 11 — Configuration of the callsign, team color and local marker color.

Operating procedure

29. Enter the callsign assigned for the mission.
30. Select the team color specified by the operational plan.
31. Choose the local marker color and press Save.
32. Verify the new identity on the map and in presence messages.

CAUTION Changing the callsign during operations may create duplicates or confusion among peers that are already connected.

4.3 GPS: position source

GPS allows the local source to be selected. In NMEA mode, the COM port and baud rate are available; the screen also lists the detected ports.

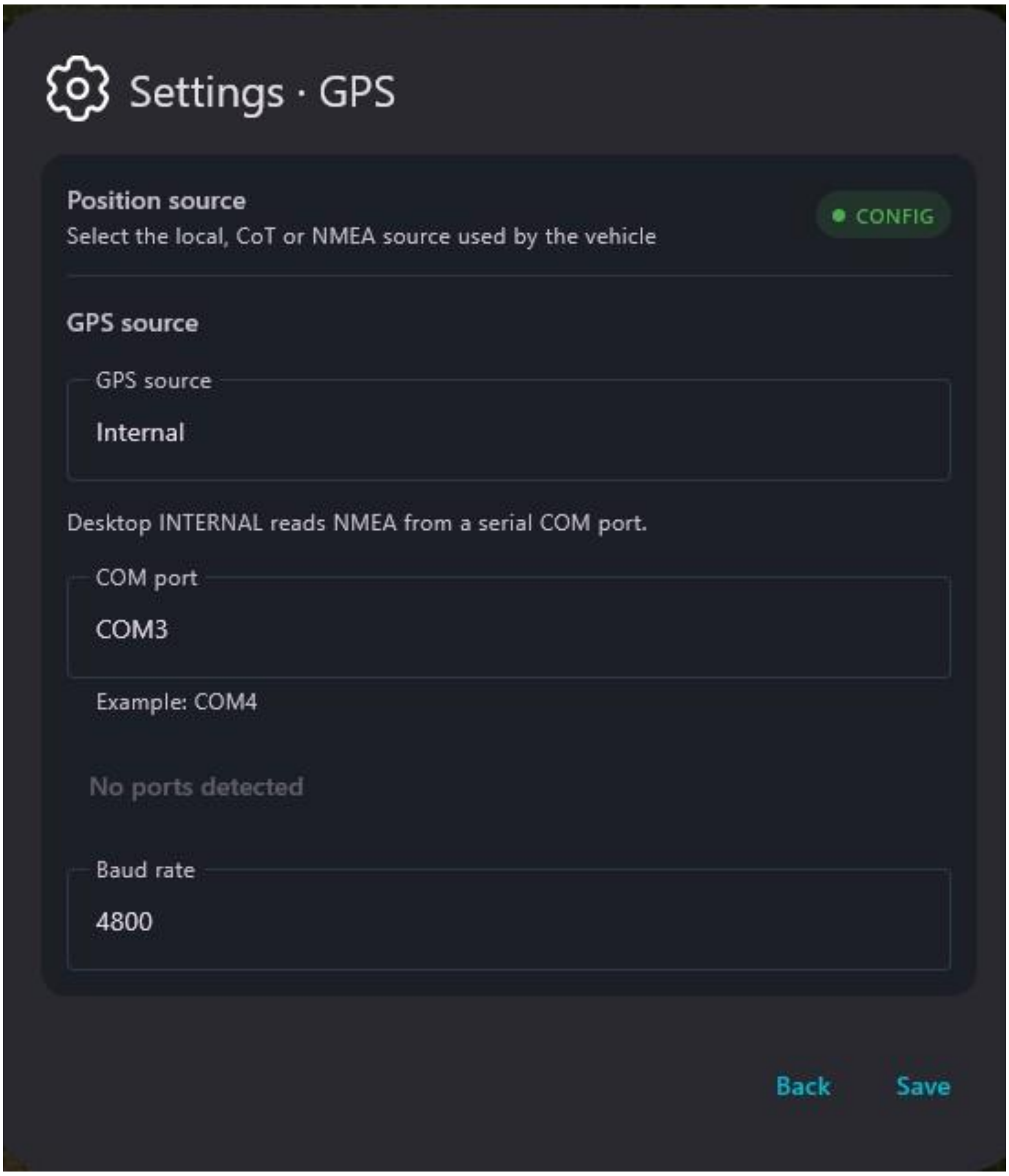

Figure 12 — GPS settings with internal or NMEA source and serial parameters.

Operating procedure

33. Select Internal when using the device position.
34. For NMEA, select the correct COM port and set the required baud rate.
35. Save and verify the position update in My position.

CAUTION An incorrect port or baud rate may prevent NMEA reception without producing a valid position.

4.4 Area of Interest Radar

The compact radar can be enabled or disabled and configured using a radius expressed in the displayed unit.

Figure 13 — Configuration of the compact situational awareness radar.

Operating procedure

36. Enable the radar when the circular summary is useful.
37. Adjust the radius according to the operational scale.
38. Save and verify the number of entities represented in the radar.

4.5 Presence

Presence controls how often the device announces its identity and position to other nodes.

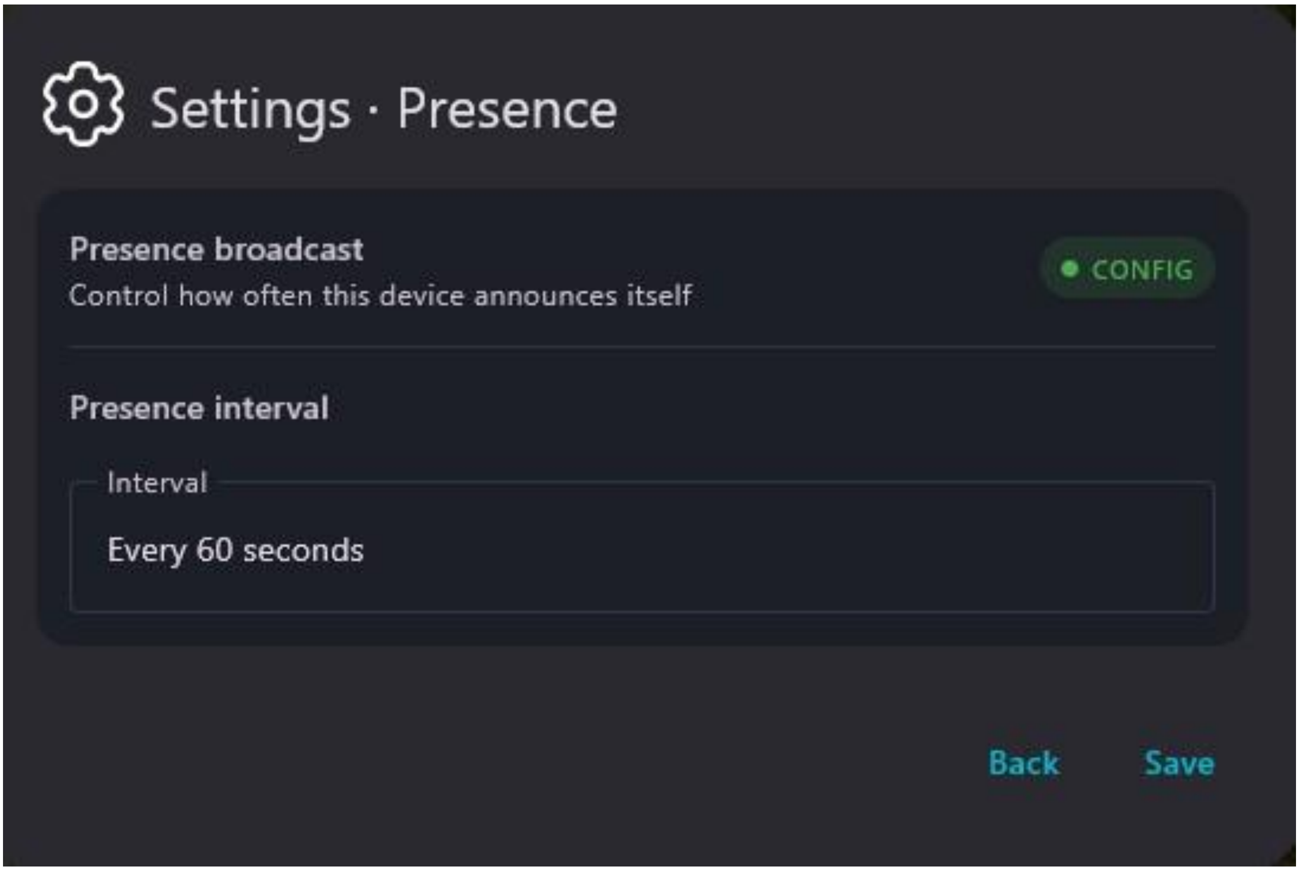

Figure 14 — TAK presence transmission interval.

Operating procedure

39. Select the interval required by the network profile.
40. Use longer intervals when bandwidth is limited and the procedure permits it.
41. Save and check the transmitted traffic on the Network dashboard.

CAUTION Very short intervals increase traffic and may be unsuitable for narrowband networks.

4.6 DTED

DTED allows a folder or archive containing elevation tiles used by Slope and other terrain analysis tools to be specified.

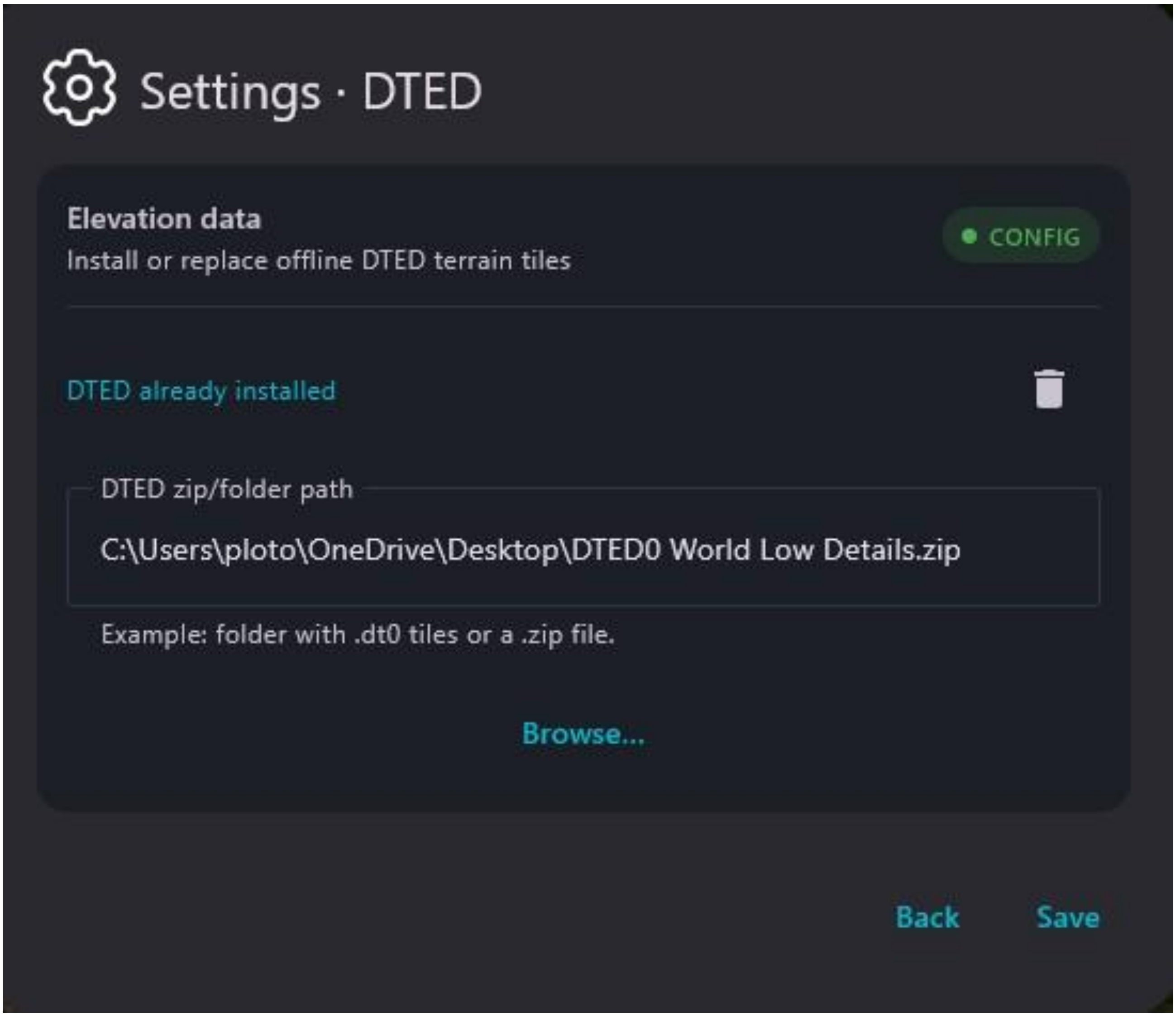

Figure 15 — Import and management of DTED elevation data.

Operating procedure

42. Press Browse and select the required folder or ZIP file.
43. Save the configuration.
44. Open Slope in a covered area to verify that the data is available.

OPERATIONAL NOTE Profile quality depends on the coverage, resolution and integrity of the dataset.

4.7 Crypto

Crypto manages message protection using a shared group key. The screen displays the CONFIG or Disabled status.

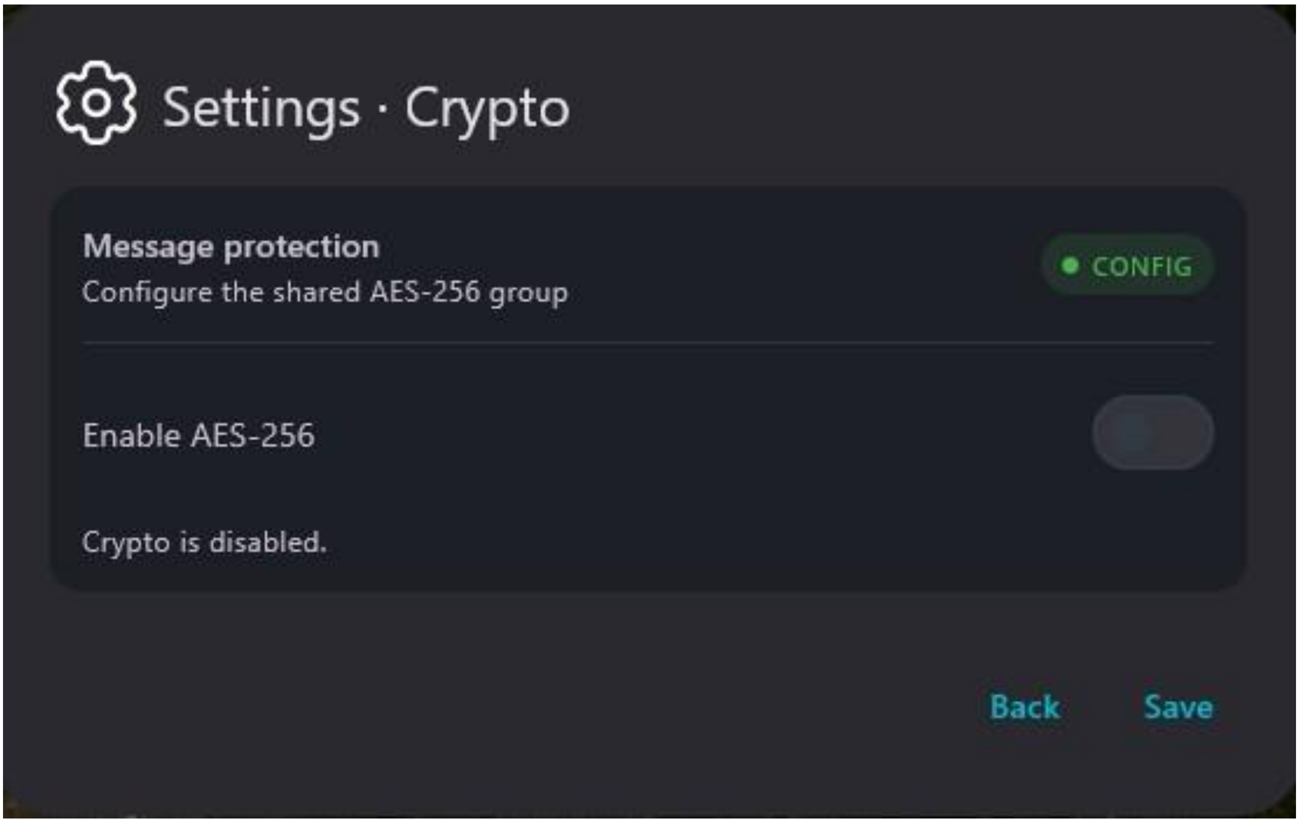

Figure 16 — AES-256 group protection configuration.

Operating procedure

45. Enable encryption only with cryptographic material distributed through authorized means.
46. Verify that all group nodes use the same configuration.
47. Save and perform a controlled transmission and reception test.

CAUTION Do not enter or transmit keys, passwords or cryptographic material through unauthorized channels.

4.8 Keyboard

Keyboard enables the virtual keyboard designed for touchscreen use. Read-only fields and drop-down menus do not activate the keyboard.

Figure 17 — Windows virtual keyboard configuration.

Operating procedure

48. Enable the function on touchscreen terminals without a physical keyboard.
49. Test an editable text field.
50. Disable it when it interferes with the use of a hardware keyboard.

4.9 Photo

Photo selects the compression profile for images associated with the map. The Radio profile produces smaller files for bandwidth-constrained links.

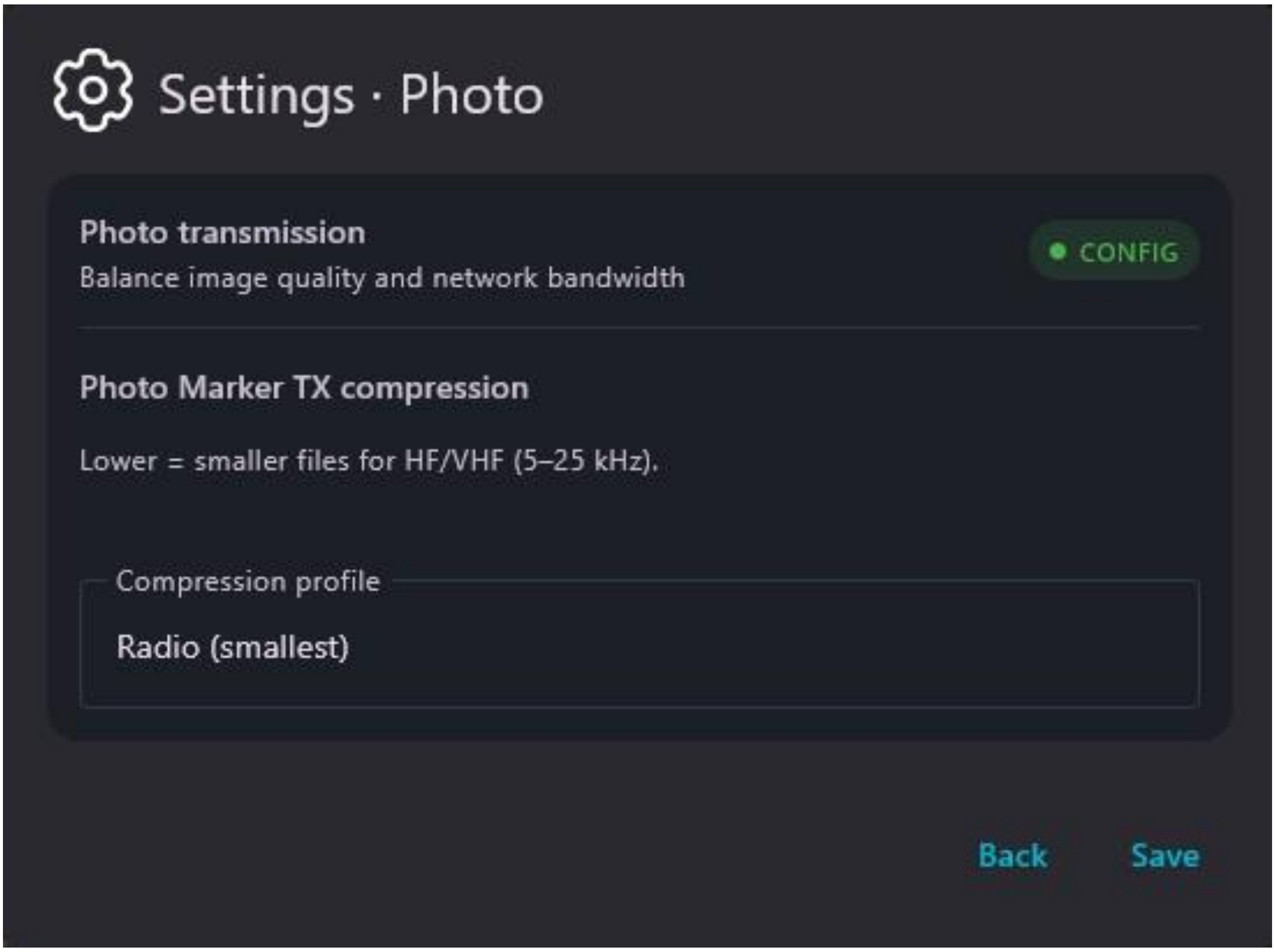

Figure 18 — Compression profile for Photo Marker transmission.

Operating procedure

51. Select the profile appropriate to the link capacity.
52. Save the configuration.
53. Send a test image and check its size and readability at the recipient.

OPERATIONAL NOTE More aggressive compression reduces file size but may limit useful detail.

4.10 Themes

Themes allows the desktop appearance to be selected and the main colors to be previewed before saving.

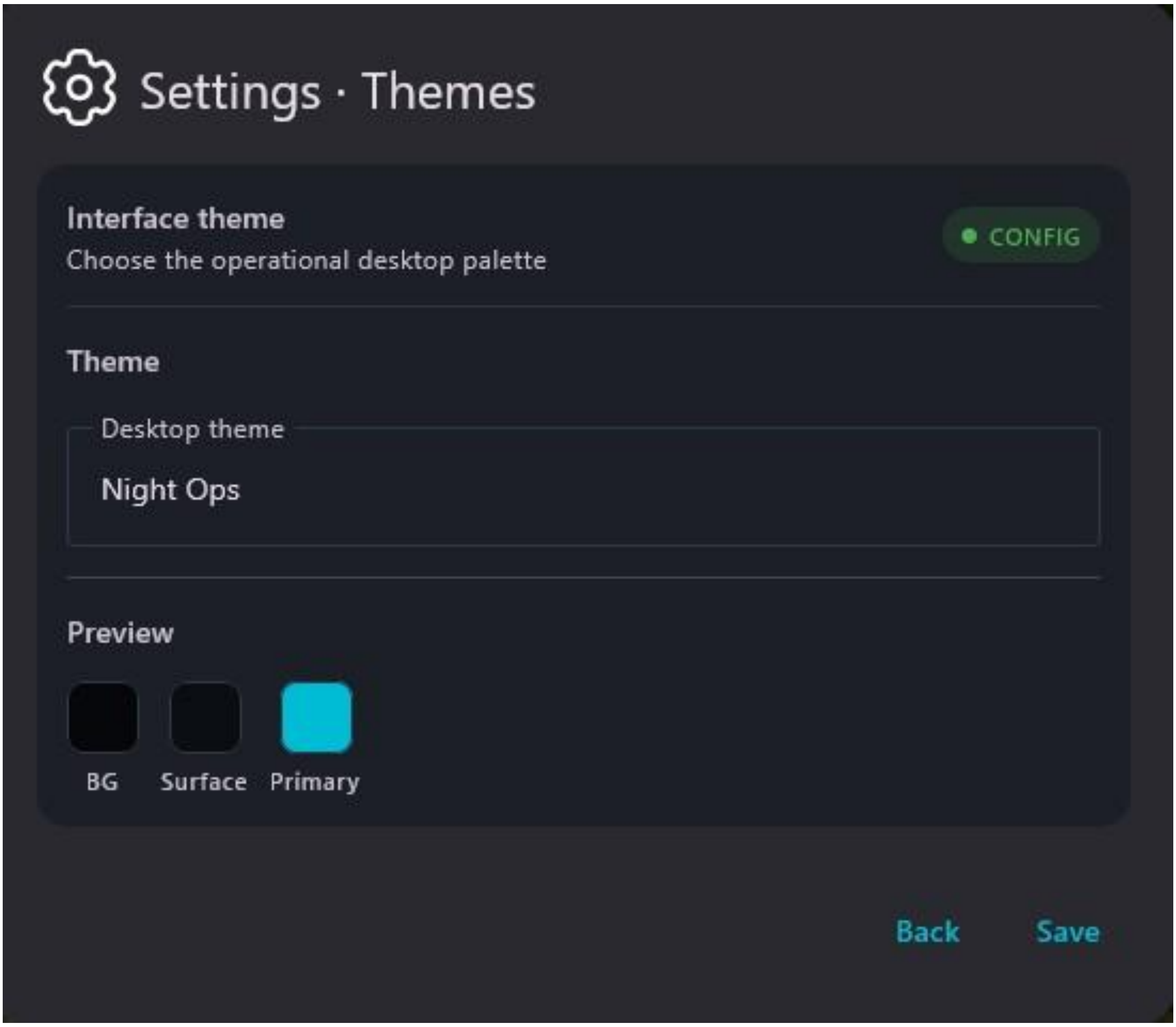

Figure 19 — Interface theme selection and palette preview.

Operating procedure

54. Select the theme appropriate to the lighting conditions and vehicle procedures.
55. Check the readability of text, icons and symbols.
56. Press Save and reopen a panel to verify the new palette.

4.11 Logs

Logs enables recording of incoming and outgoing CoT messages and shows the file path. A command is available to open the log folder.

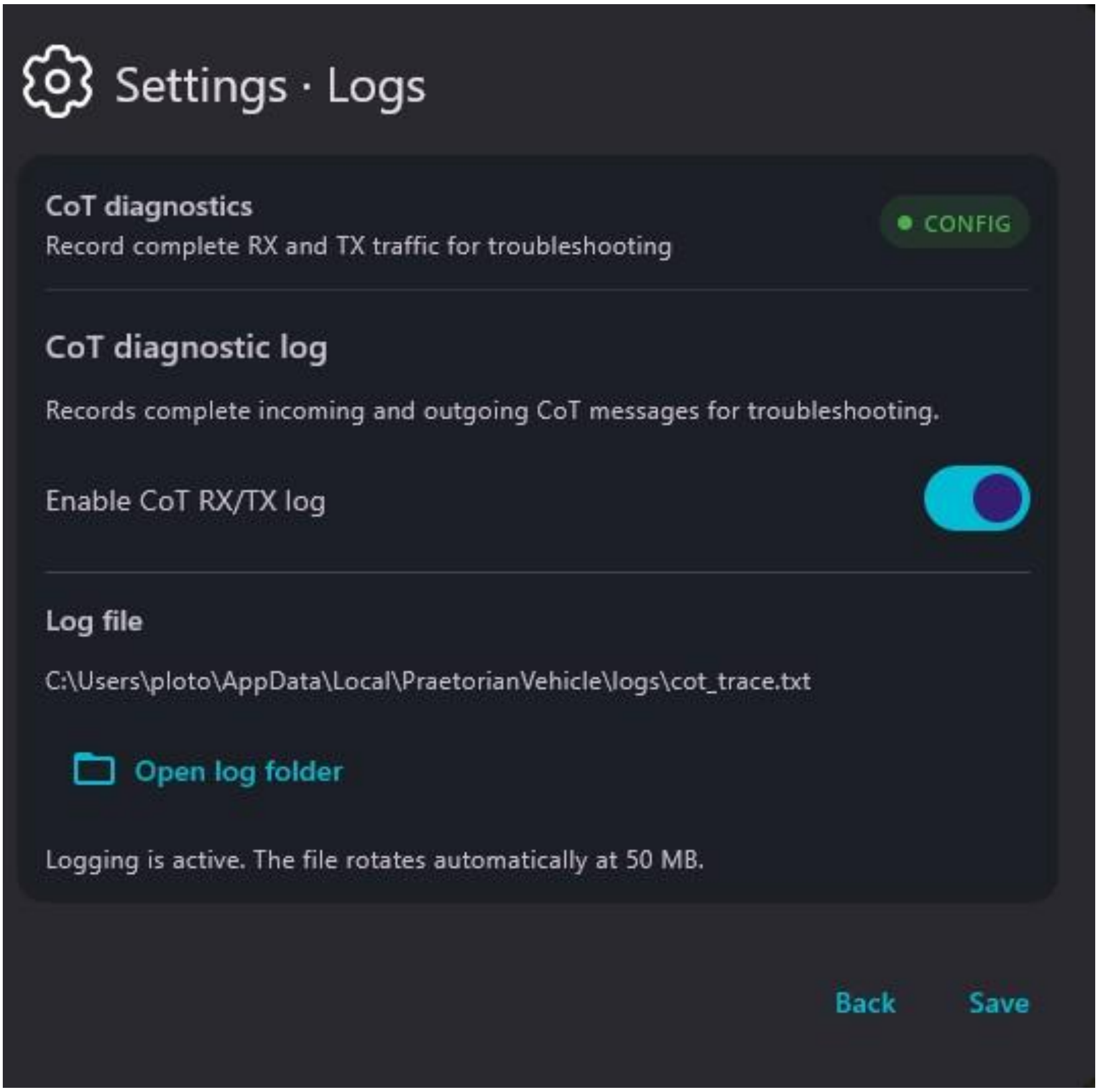

Figure 20 — Complete diagnostic logging of CoT RX/TX traffic.

Operating procedure

57. Enable logs only for diagnostics or as directed by technical personnel.
58. Reproduce the anomaly while noting the time.
59. Open the folder and provide only the authorized file.
60. Disable logging when it is no longer required.

CAUTION Logs may contain addresses, identifiers, coordinates and operational data. Handle them in accordance with the applicable security rules.

5. Maps, terrain and elevation data

The Maps module manages offline map bases and online sources. Heatmap and Slope provide additional terrain interpretation tools.

5.1 Offline maps

The Offline tab shows the maps imported on the device and the commands to add, refresh, open the folder or delete packages. The ZIP file must comply with the structure indicated by the interface.

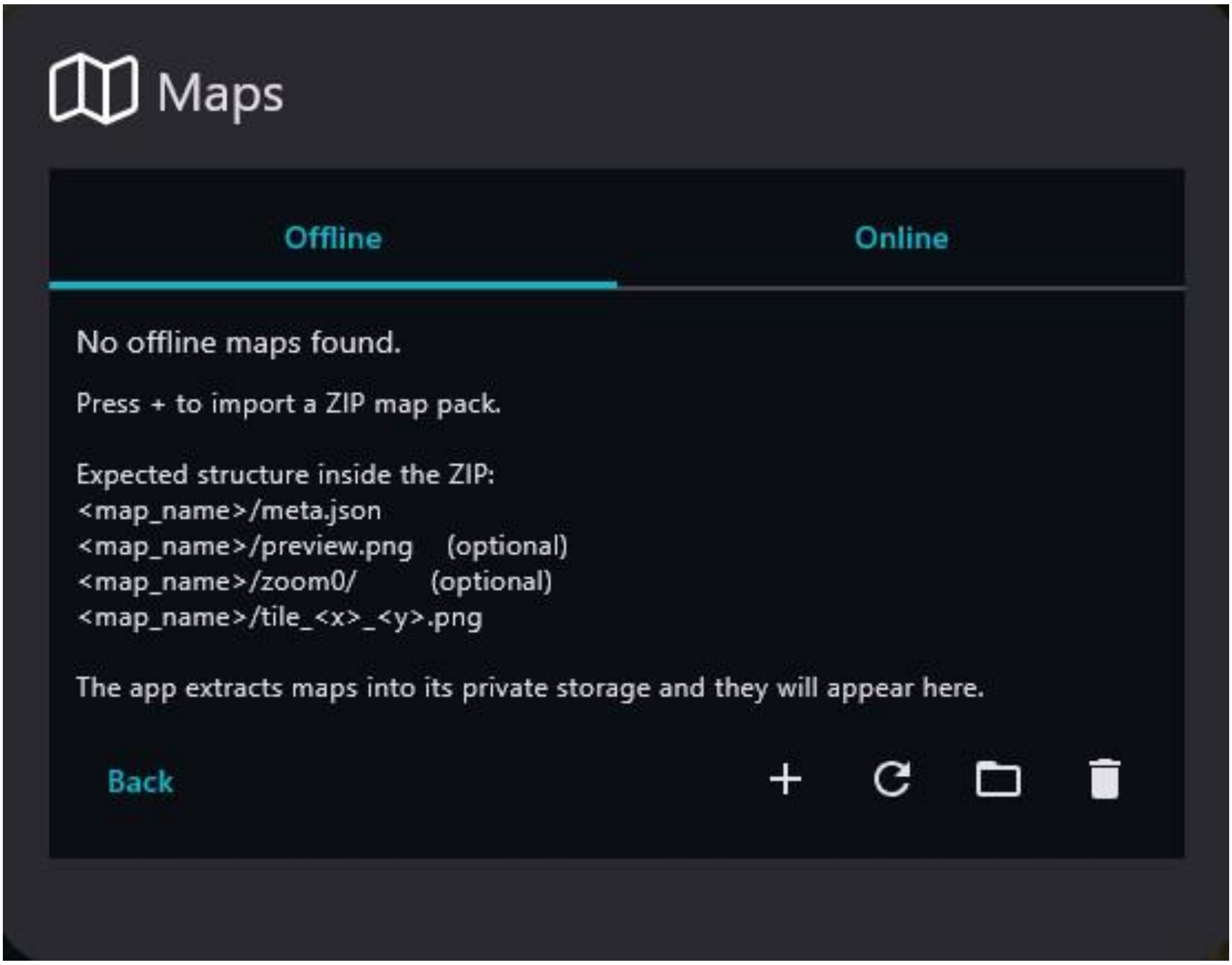

Figure 21 — Offline tab with instructions and commands for importing a ZIP map package.

Operating procedure

61. Press + and select the authorized ZIP package.
62. Wait for extraction and verify that the map appears in the list.
63. Enable the map and check zoom, metadata and coverage.
64. Delete a package only when it is no longer required.

OPERATIONAL NOTE Do not manually modify the internal package structure during import.

5.2 Online map sources

The Online tab presents the available sources and allows the one used as the COP base map to be selected. Actual availability depends on connectivity and the service terms.

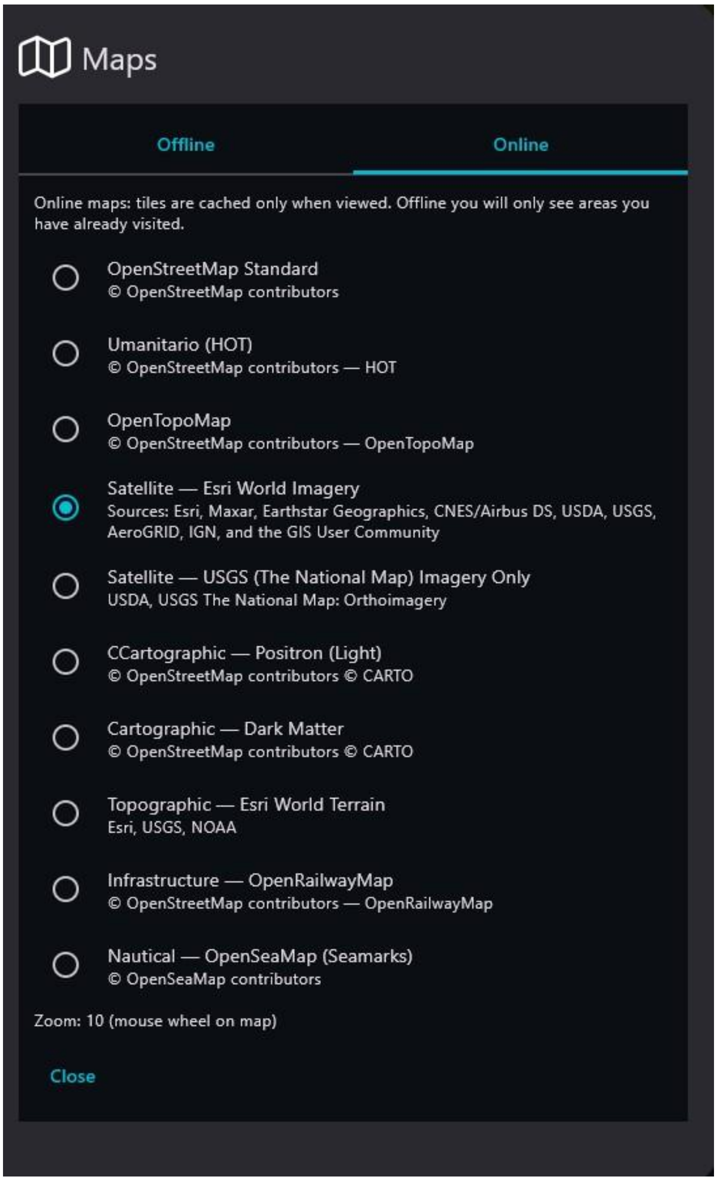

Figure 22 — List of selectable online map sources.

Operating procedure

65. Select a single source appropriate to the area and mission.
66. Verify that tiles are available at the required zoom level.
67. Switch to an offline map when the network cannot guarantee continuity.

OPERATIONAL NOTE Online maps must not be considered available without a connection or on isolated networks.

5.3 Heatmap

Heatmap overlays thematic coloring on the map to highlight terrain variations over large areas.

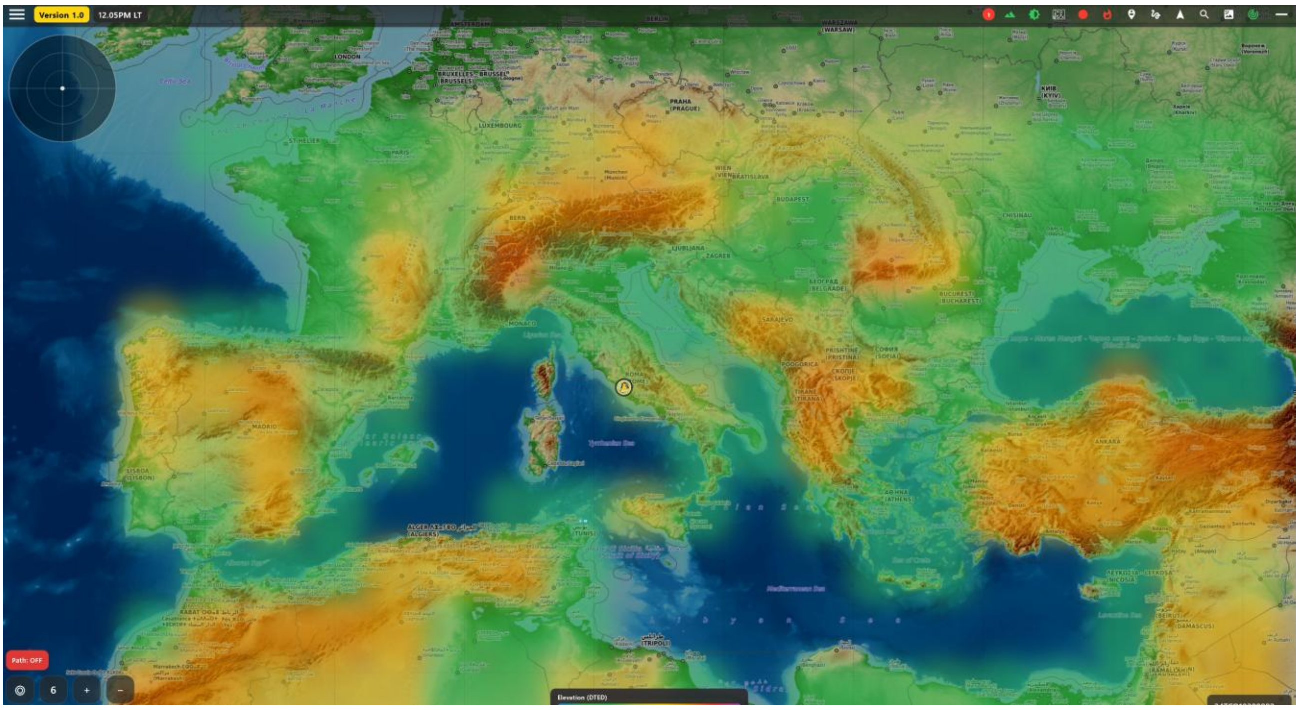

Figure 23 — Heatmap view applied to the base map.

Operating procedure

68. Enable the layer and check the associated legend.
69. Compare the color interpretation with the base map and elevation data.
70. Disable the layer if it reduces symbol readability.

OPERATIONAL NOTE The meaning of the colors depends on the source and layer configuration; always check the legend.

6. Communications, contacts and Data Packages

Chat, Contacts, Alert, FORM MSG and Package cover free-text communications, rapid messages, structured reports and file exchange.

6.1 Chat conversation list

The Conversations tab shows existing conversations, the latest message and the time. The available icons allow a local conversation to be opened or removed.

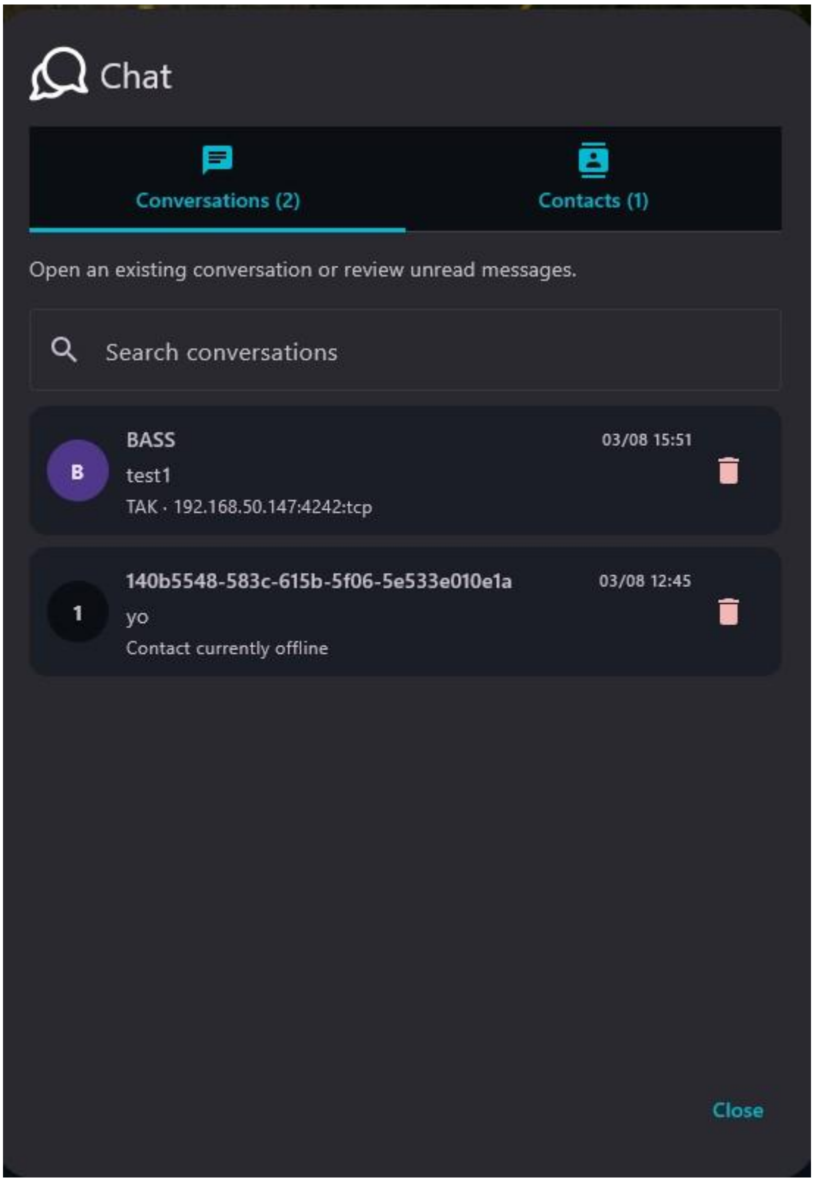

Figure 24 — Conversations tab with open conversations and received messages.

Operating procedure

71. Select the correct conversation.
72. Check the identifier and latest message before replying.
73. Remove the local history only when required and in accordance with procedure.

6.2 Chat contacts

The Contacts tab lists the available compatible peers. Search allows the recipient to be filtered quickly.

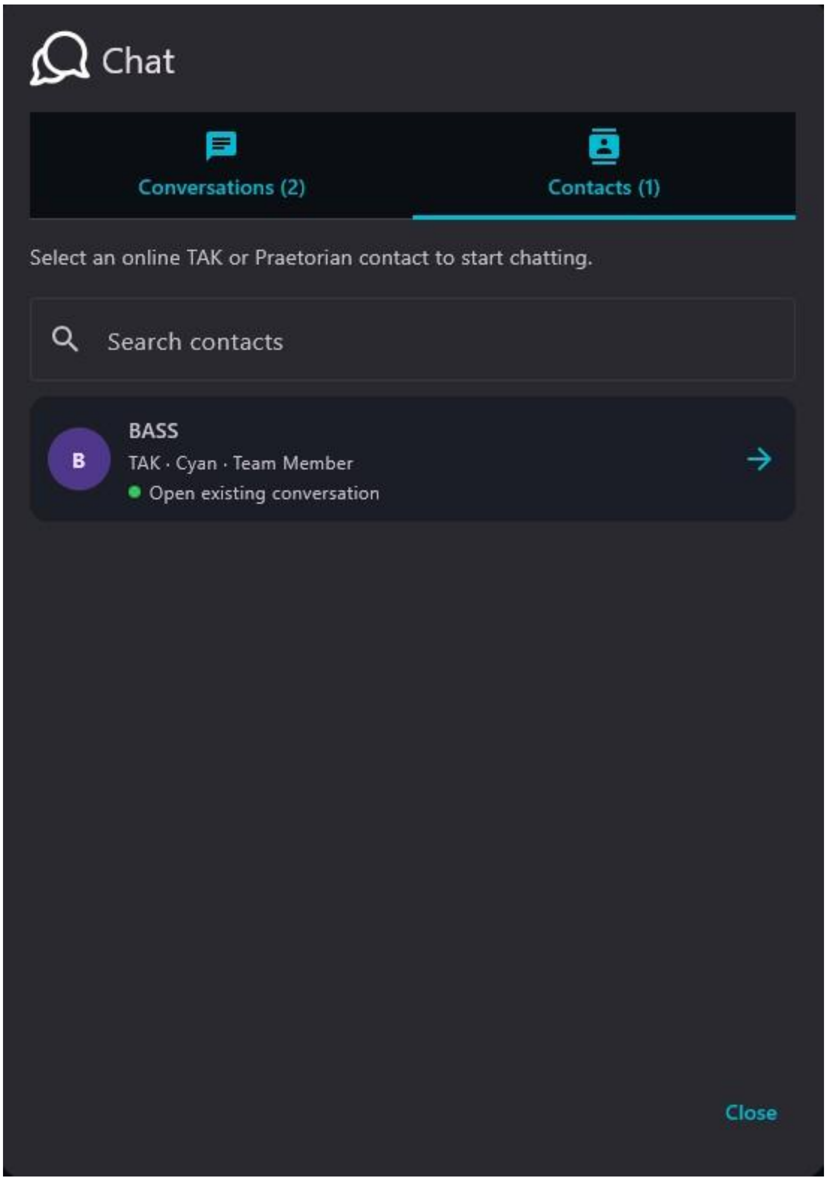

Figure 25 — Contacts tab with search and start of a new conversation.

Operating procedure

74. Search for the expected callsign or identifier.
75. Verify that the peer is the authorized one.
76. Open the conversation using the side arrow.

6.3 Direct chat on the COP

The conversation remains overlaid on the COP to preserve the geographic context. Sent and received messages are visually distinguished and include the time.

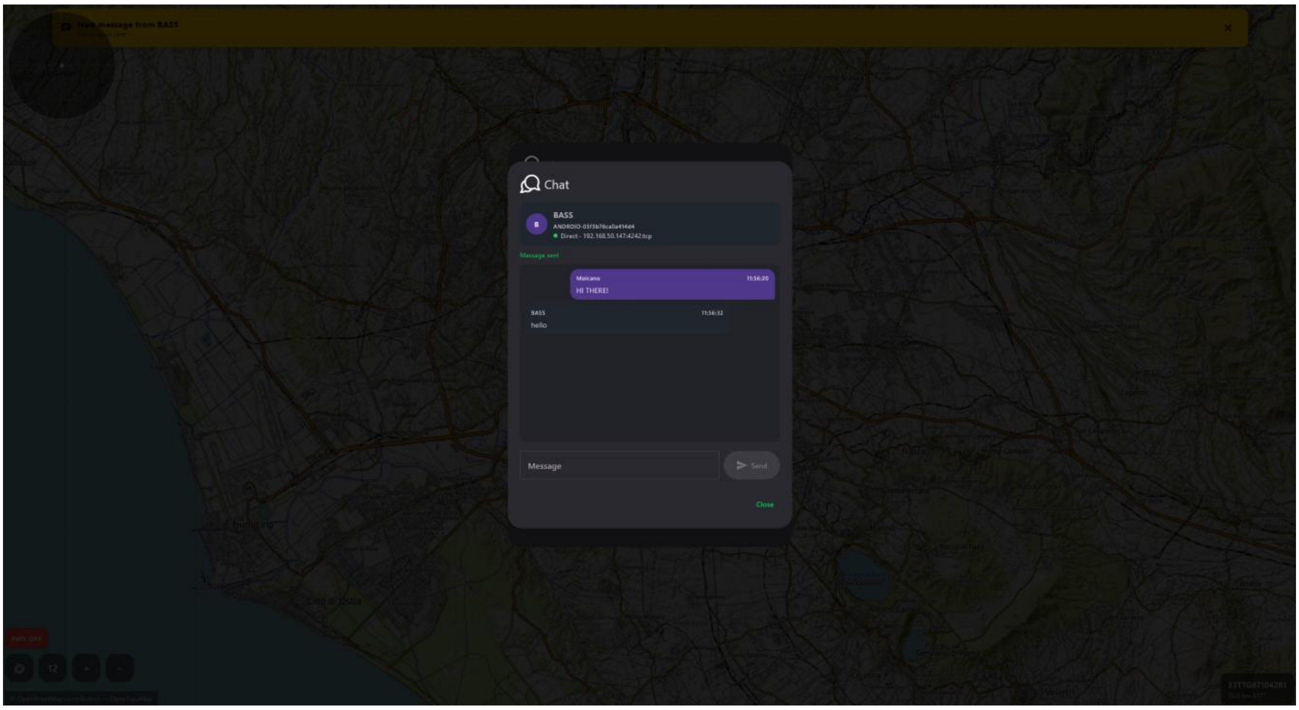

Figure 26 — Direct chat window open over the map.

Operating procedure

77. Verify the callsign, identifier and connection status.
78. Enter a short, unambiguous message.
79. Press Send once and wait for the result or reply.

OPERATIONAL NOTE Technical confirmation of transmission does not constitute confirmation of reading.

6.4 Contacts

Contacts shows compatible peers and their status. The Package command opens the file-sending workflow for the selected contact.

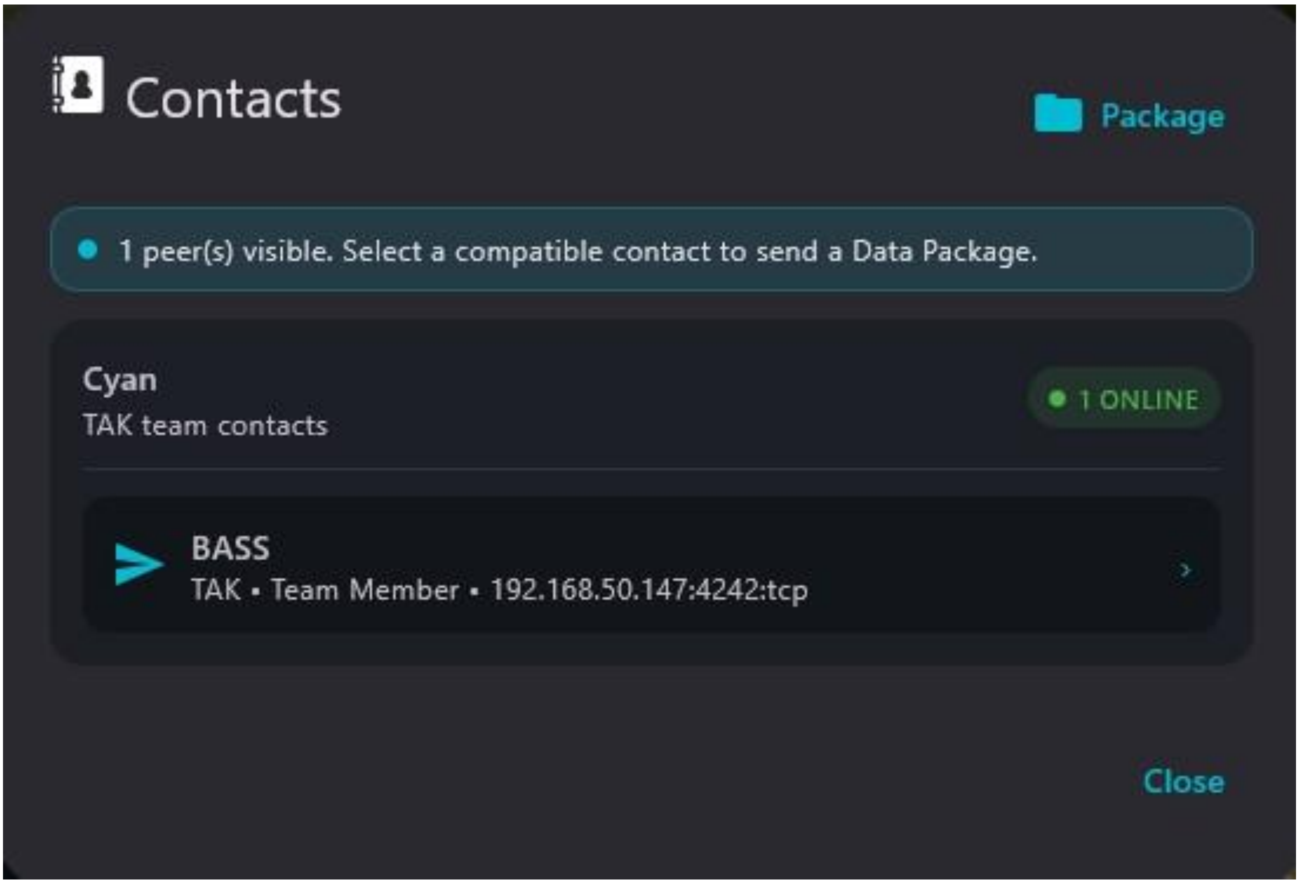

Figure 27 — List of visible TAK peers and access to Data Package transmission.

Operating procedure

80. Verify the number of visible peers.
81. Select the contact and check the callsign, role and address.
82. Open Package only after confirming the recipient.

6.5 Sending a Data Package to a contact

The panel shows the selected recipient, the network target and the READY status. The file is selected with Choose file.

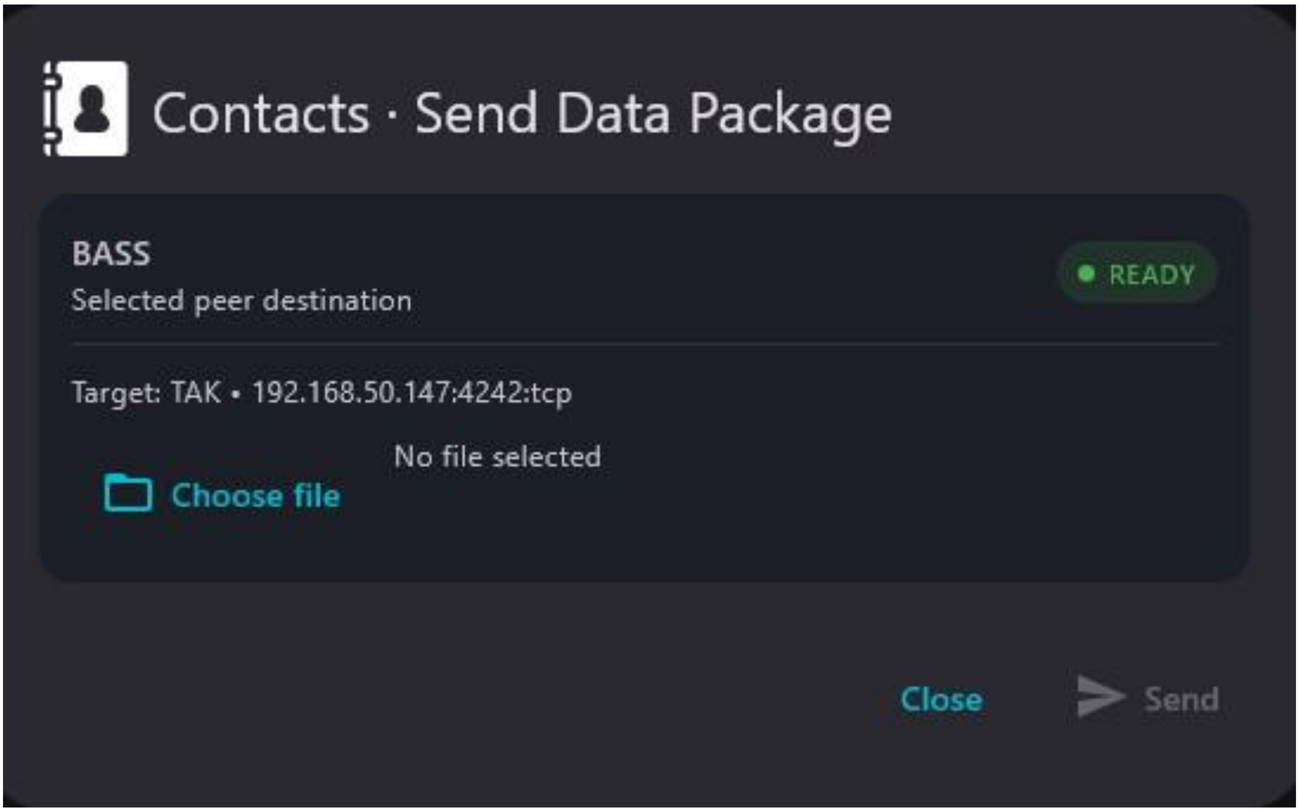

Figure 28 — Data Package transmission panel with destination and file selection.

Operating procedure

83. Check the displayed recipient.
84. Press Choose file and select the authorized package.
85. Verify the file name and size.
86. Press Send and wait for confirmation.

CAUTION Do not send files that have not been verified, are classified or are intended for another network.

6.6 Alert presets

Alert presets allows a short name and text to be assigned to four shortcuts. In the example, MEDIC, IED, CONTACT and VEHICLE are configured.

Figure 29 — Configuration of the four rapid Alert messages.

Operating procedure

87. Enter a unique designation in Alert name.
88. Enter the standard text in Message sent.
89. Press Save and perform a controlled test.
90. Always confirm the recipient before operational use.

CAUTION Presets must be agreed by the organization; avoid non-standard abbreviations or text that could be interpreted ambiguously.

6.7 FORM MSG catalog

FORM MSG includes templates such as 9 LINE, CALL FOR FIRE, CONTACT, IED, AIR SUPPORT, SITREP, VEHICLE REPORT, PERSON / INTERPRETER / DETAINEE REPORT and THREAT.

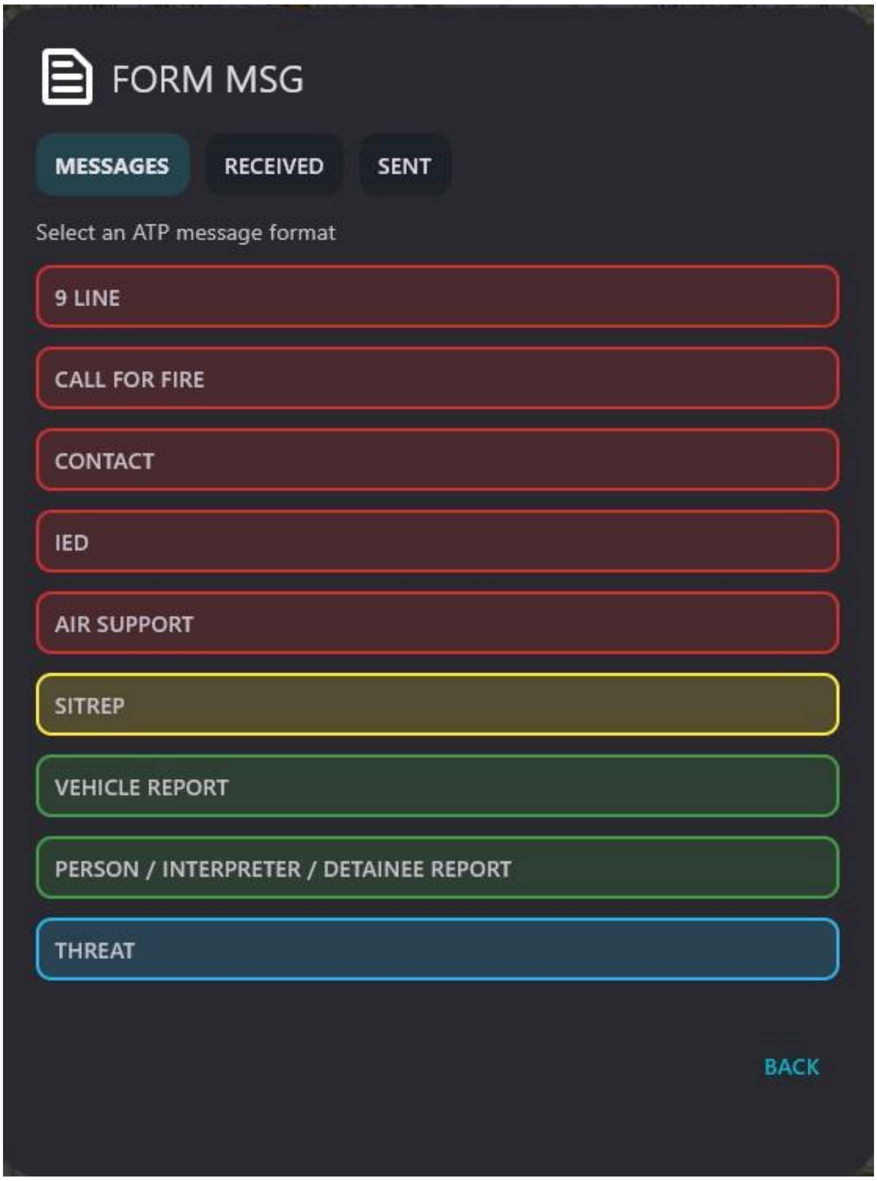

Figure 30 — Catalog of structured message templates.

Operating procedure

91. Open the Messages tab.
92. Select the format required by the procedure.
93. Complete all mandatory fields and verify the coordinates and time.
94. Use Save for a draft or Send for transmission.

6.8 Completing a SITREP

In the SITREP template, the date/time and MGRS position may be prefilled. The operator completes the description of the event and the actions in progress.

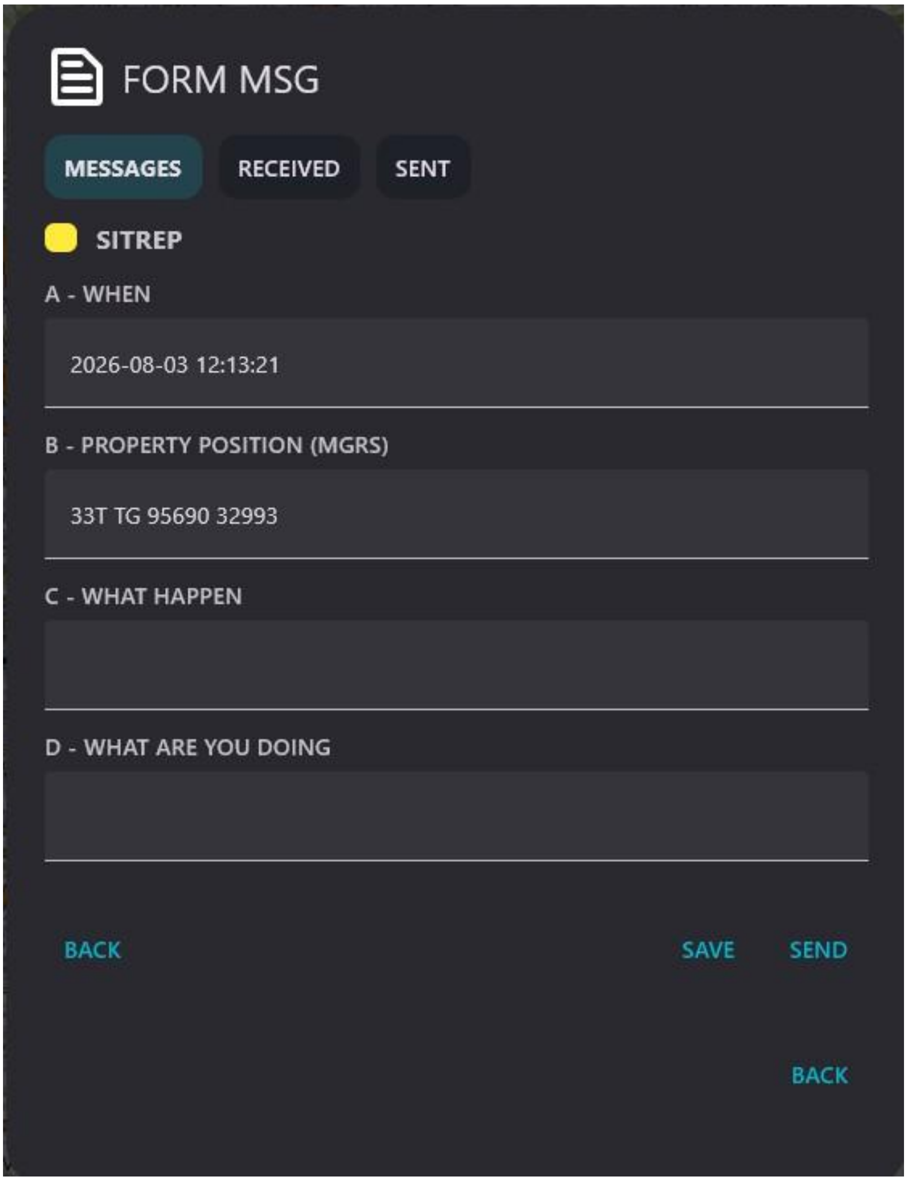

Figure 31 — SITREP form with date/time, position, event and current activity.

Operating procedure

95. Check A - WHEN.
96. Verify B - PROPERTY POSITION (MGRS).
97. Complete C - WHAT HAPPEN with a concise description.
98. Complete D - WHAT ARE YOU DOING.
99. Save or send after the final check.

6.9 Data Package management

Package separates files received from peers, files already downloaded to the device and files available in the TAK Server repository. Query Server, Upload, Refresh and Open folder are available.

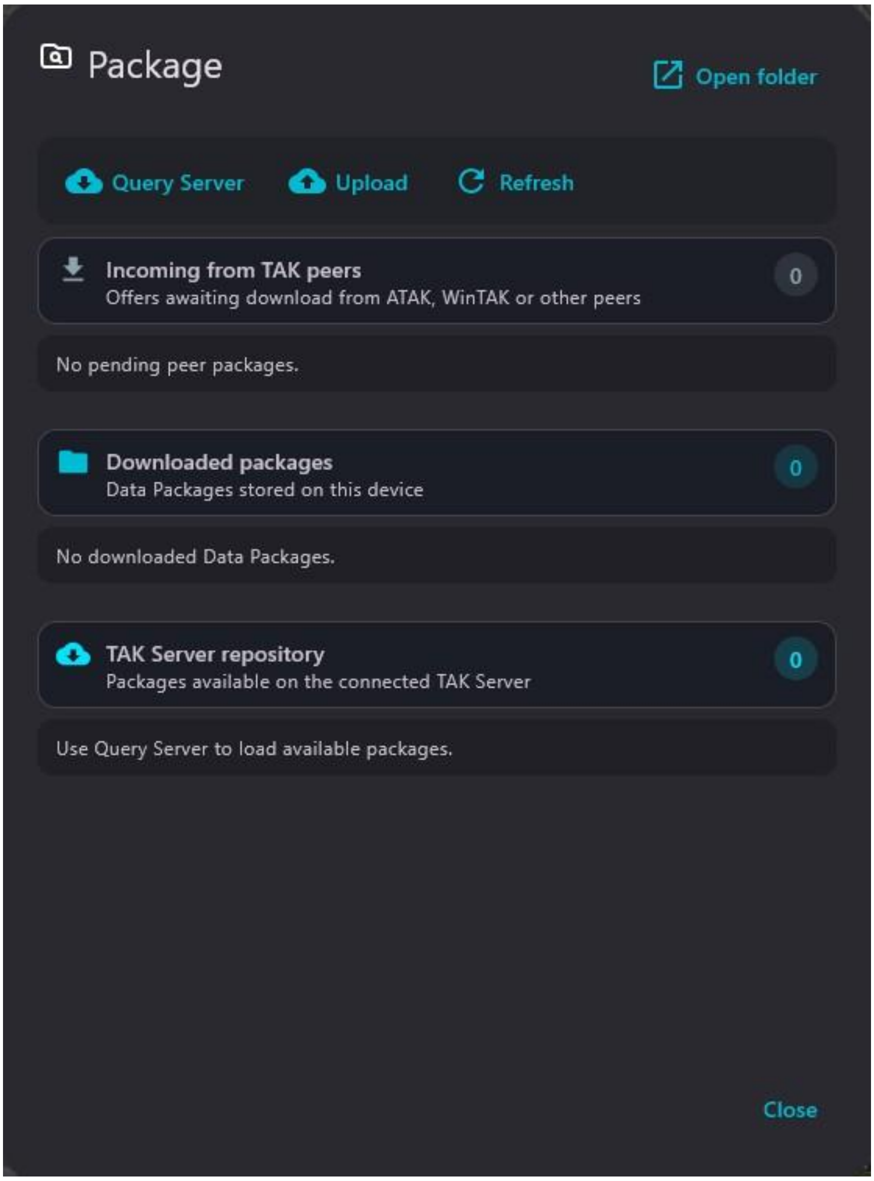

Figure 32 — Package panel with incoming packages, local packages and the TAK Server repository.

Item	Purpose	Operating guidance
Query Server	Queries the connected TAK Server.	Requires valid connectivity and authorization.
Upload	Uploads a package to the repository or the intended workflow.	Verify the destination and content.
Refresh	Refreshes the three areas of the panel.	Use it after downloads, uploads or receptions.
Open folder	Opens the local package folder.	Do not modify files in use by the application.

7. Network, LEGIO v2 and RADIO SA

Network is the central point for checking traffic and configuring interfaces, listeners, destinations, servers and narrowband profiles.

7.1 Network dashboard

The upper section shows TAK Server status and the Received and Transmitted counters. The lower cards open My IP, Input, Output, Server 1, Server 2 and LEGIO v2.

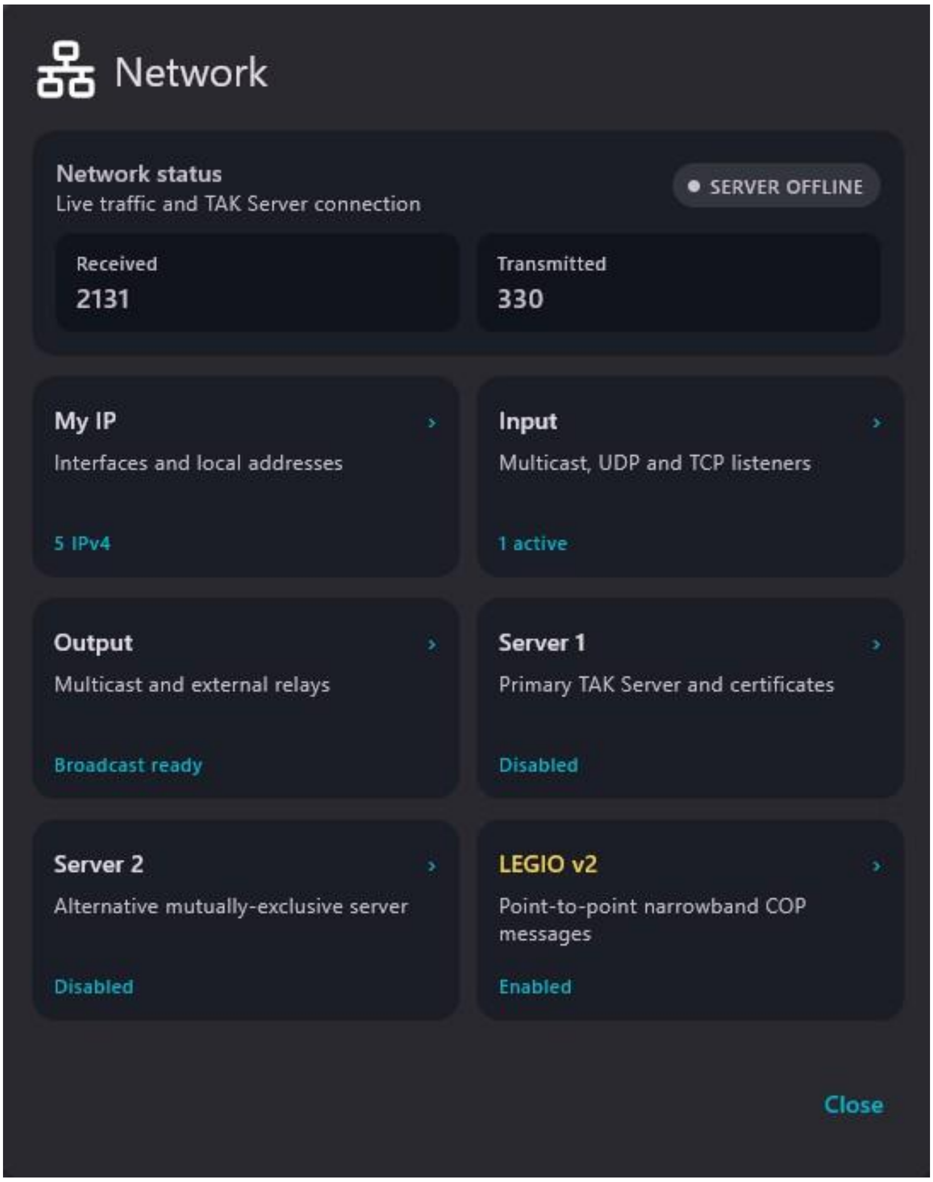

Figure 33 — Network dashboard with status, counters and configuration cards.

Operating procedure

100. Check the overall status and counters.
101. Open My IP to identify the operational interface.
102. Verify Input and Output according to the authorized architecture.
103. Open the server or LEGIO only when required.

7.2 My IP

My IP lists physical and virtual adapters with IPv4/IPv6 addresses. Refresh interfaces updates the list.

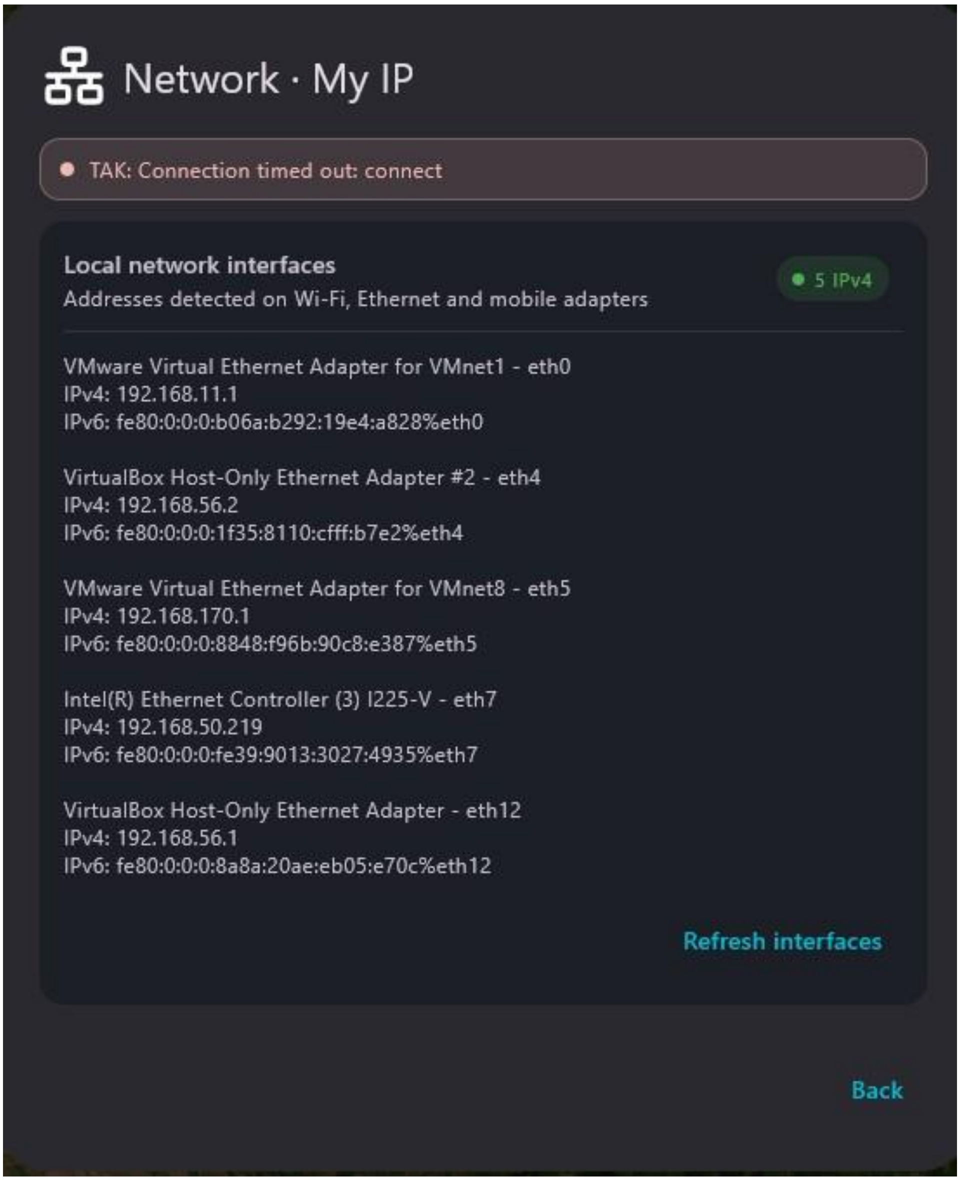

Figure 34 — List of local interfaces and detected addresses.

Operating procedure

104. Identify the interface connected to the tactical network.
105. Record the IPv4 address specified by the network plan.
106. Use Refresh after a connection or configuration change.

OPERATIONAL NOTE Virtual adapters, VPNs and bridges may be present but may not correspond to the operational network.

7.3 Input

Input defines the traffic reception points. Each listener provides an enable switch, interface or address, and port.

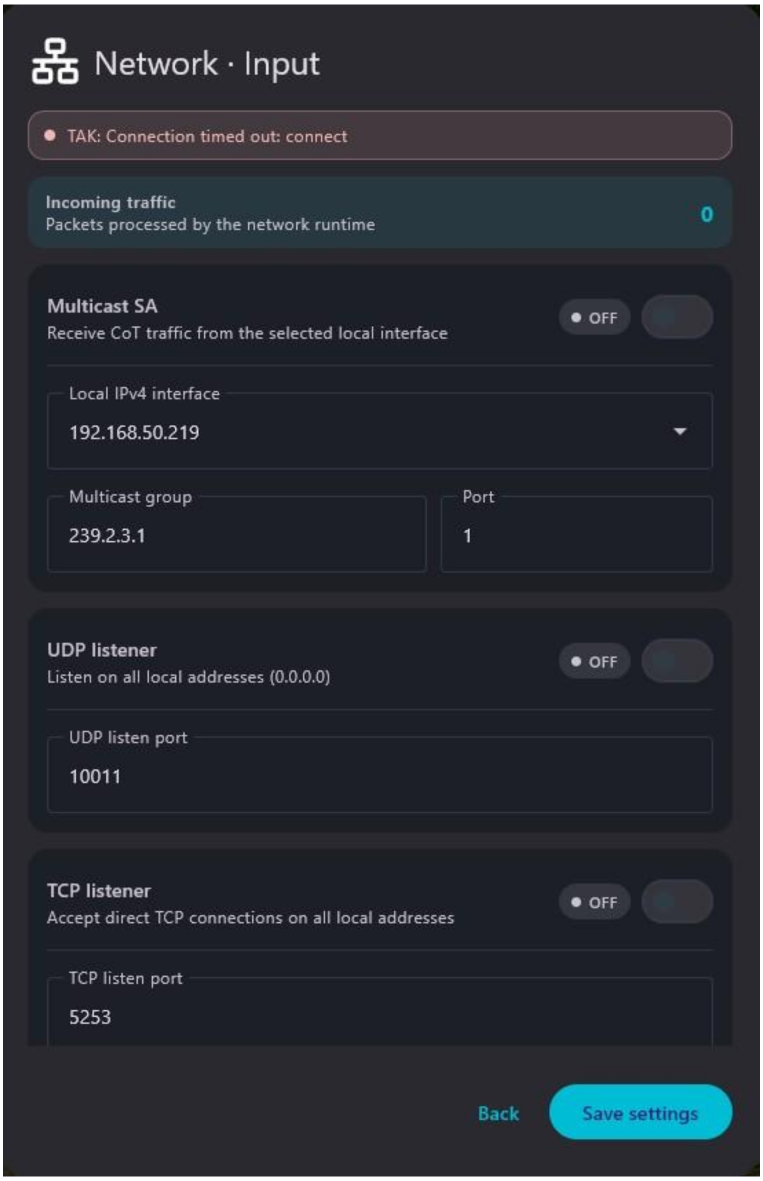

Figure 35 — Configuration of multicast, UDP and TCP listeners.

Operating procedure

107. Enable only the required listeners.
108. Select the correct local interface for multicast.
109. Enter UDP/TCP ports that match the sources.
110. Press Save settings and verify that Received increases.

CAUTION Duplicate ports or local firewalls may prevent the listener from opening.

7.4 Output

Output defines where Praetorian transmits data. The screen includes SA multicast, NRT relay or other configured outputs, with interface, address and port.

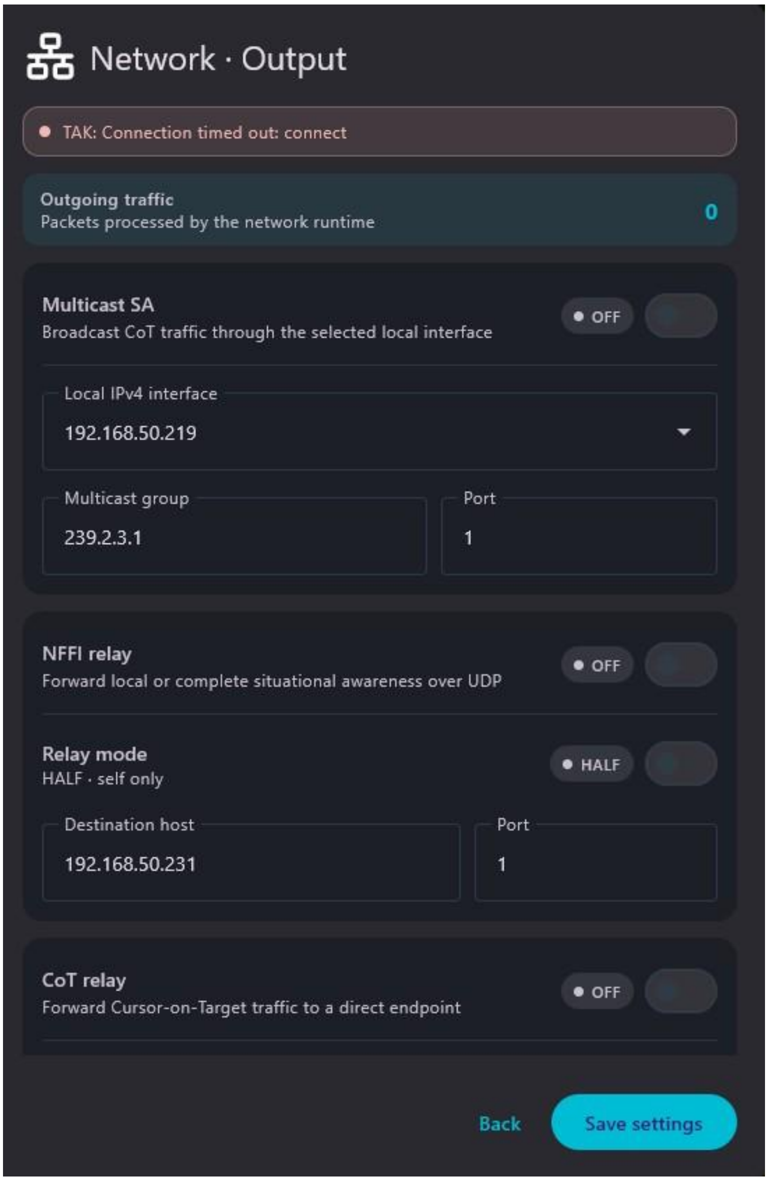

Figure 36 — Configuration of multicast destinations and external relays.

Operating procedure

111. Enable only the outputs specified by the network plan.
112. Check the address, port and interface.
113. Save and observe the Transmitted counter.
114. Verify reception on the destination system.

CAUTION Avoid circular relays that feed the same messages back into the system.

7.5 Server 1

Server 1 allows the host, port and authentication material to be set. The screen indicates connection status and allows the server to be enabled or disabled.

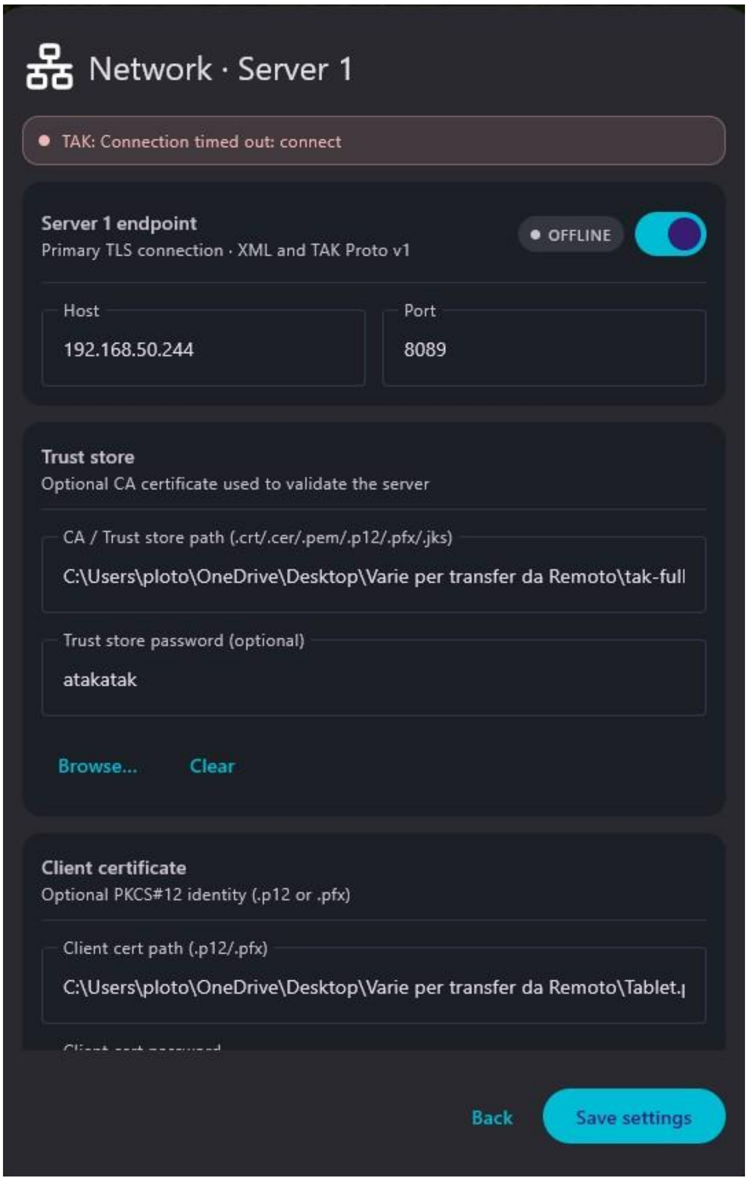

Figure 37 — Configuration of the primary TAK Server and certificates.

Operating procedure

115. Enter the host and port provided by the administrator.
116. Select the correct trust store and client certificate.
117. Save and enable the server.
118. Check the status on the Network dashboard.

CAUTION Do not share private keys, passwords or certificates through unauthorized channels.

7.6 LEGIO v2 from the Network dashboard

The LEGIO v2 screen shows any warnings, module status, narrowband profile, mission workbook and local identity.

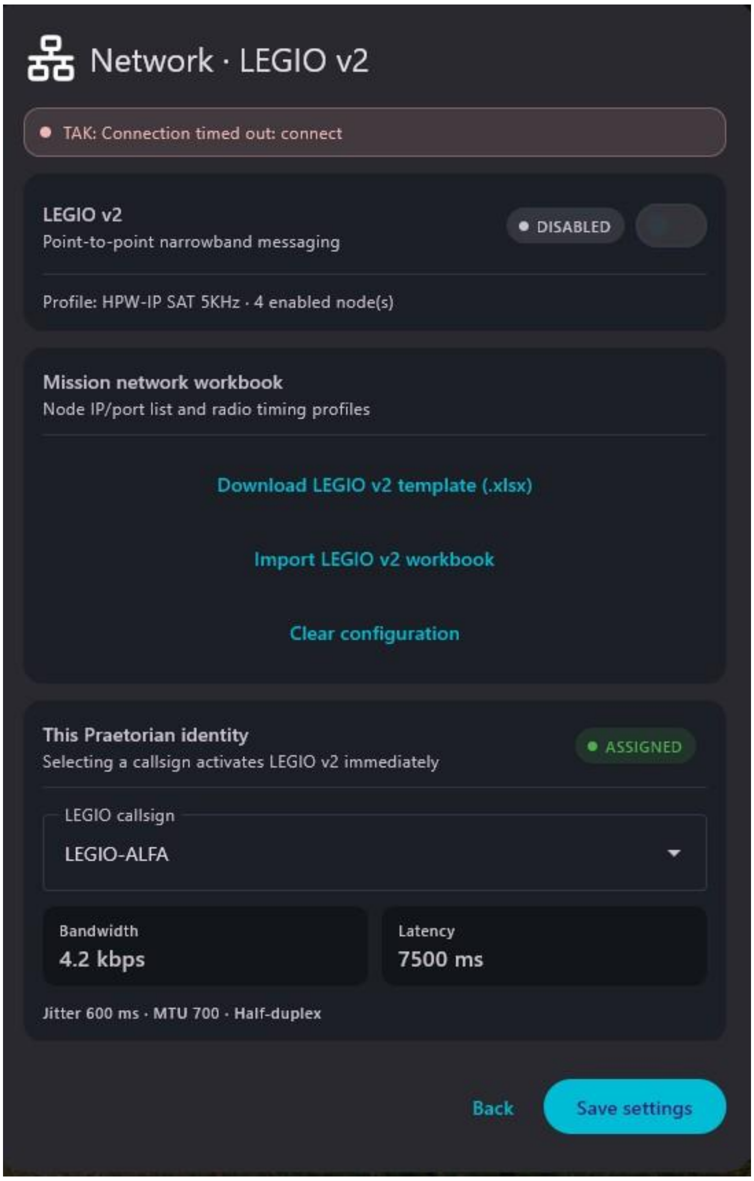

Figure 38 — LEGIO v2 panel opened from Network with a TAK connection warning.

Operating procedure

119. Read the warnings at the top.
120. Import the required mission workbook.
121. Select the local callsign and save.
122. Check the displayed bandwidth, latency, jitter and MTU.

7.7 LEGIO v2 configuration

LEGIO v2 is the point-to-point channel optimized for COP messages over narrowband tactical networks. The screen allows the template to be downloaded, the workbook to be imported and the local identity to be selected.

Figure 39 — LEGIO v2 configuration with radio profile, workbook and callsign.

Operating procedure

123. Use Download template to prepare the mission workbook.
124. Import the verified workbook.
125. Select the assigned LEGIO callsign.
126. Press Save settings and verify the ENABLED status.

CAUTION With profiles limited to a few kbit/s and high latency, prioritize short messages, COP delta and essential data.

7.8 LEGIO node list

The LEGIO interface lists the available nodes and their status. Selecting a contact opens the dedicated channel.

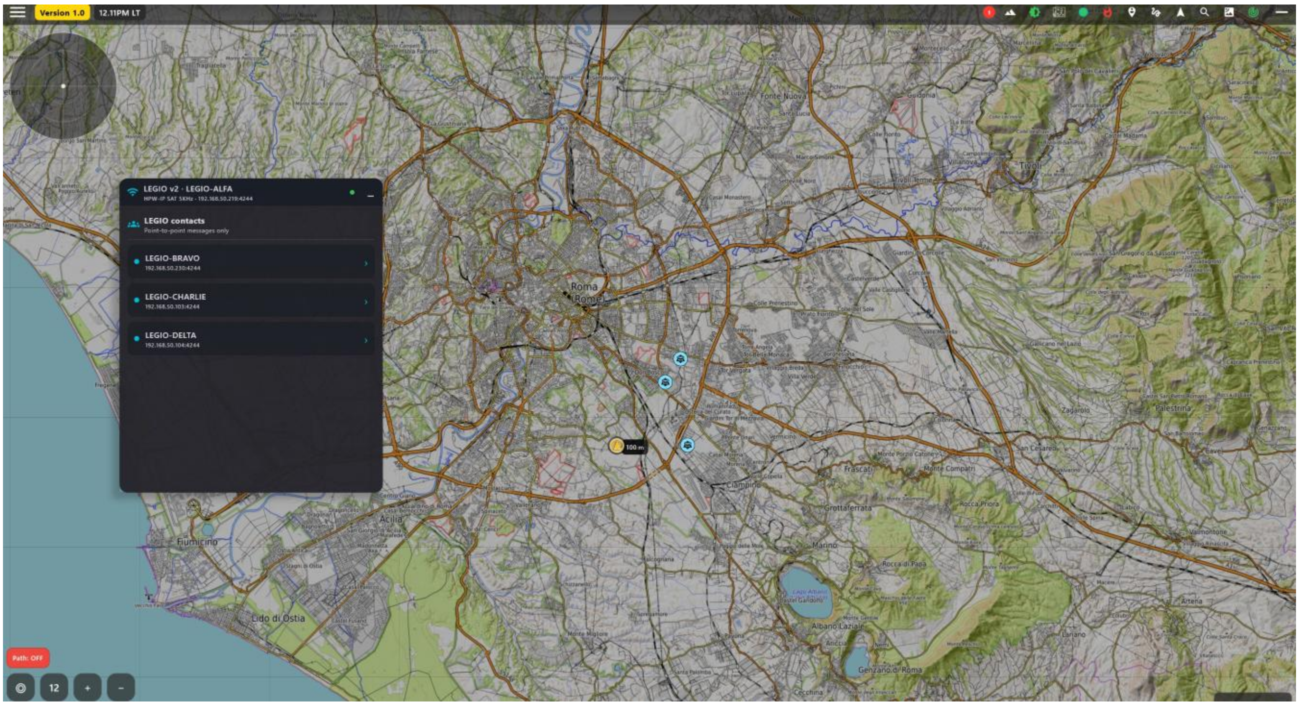

Figure 40 — List of LEGIO nodes displayed over the COP.

Operating procedure

127. Check the node status and associated profile.
128. Select the correct recipient.
129. Open the message panel and choose the content type.

7.9 LEGIO messages

The contact window provides three modes: Text for short messages, COP delta for changes only and Full COP for a complete operational picture.

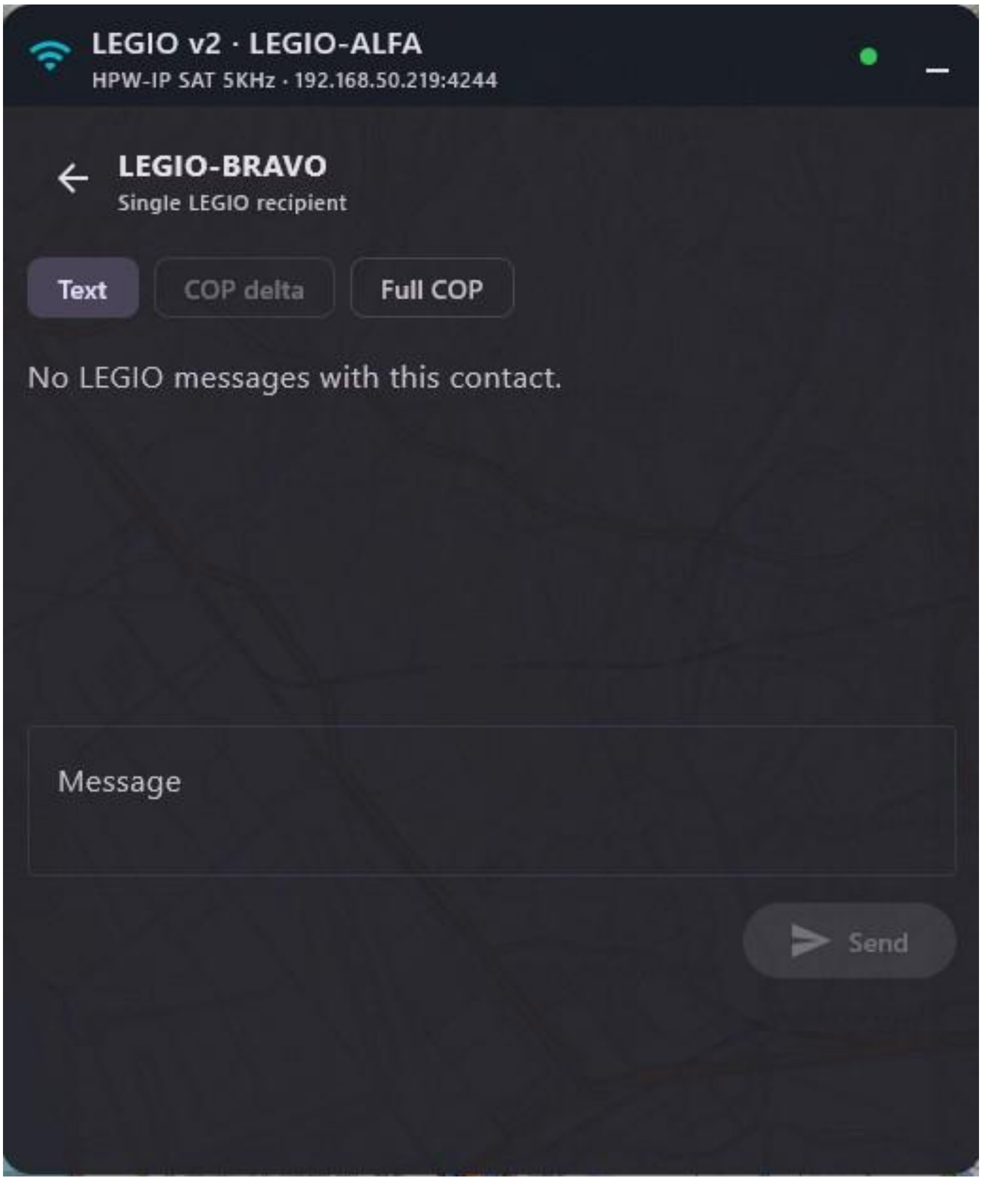

Figure 41 — LEGIO window with Text, COP delta and Full COP modes.

Operating procedure

130. Prefer Text for essential communications.
131. Use COP delta for incremental updates.
132. Use Full COP only when required and compatible with the available bandwidth.
133. Press Send once and wait for the time expected for the profile.

CAUTION High latency may delay delivery; avoid immediate retransmissions that duplicate traffic.

7.10 RADIO SA: settings

RADIO SA receives Situational Awareness messages from compatible radio sources, associates a local callsign and inserts the tracks into the COP. The main switches are Enable RADIO SA and Show assigned radios on map.

Figure 42 — RADIO SA settings and description of supported sources.

Operating procedure

134. Enable RADIO SA.
135. Verify that the required network Inputs are active.
136. Enable Show assigned radios on map to limit the display to assigned radios.
137. Check the tracks on the COP.

OPERATIONAL NOTE Praetorian applies source-aware forwarding rules; do not create circular external relays.

7.11 RADIO SA: radio list

The Radios tab shows detected sources and allows saved callsigns to be assigned. Clear aliases removes existing associations.

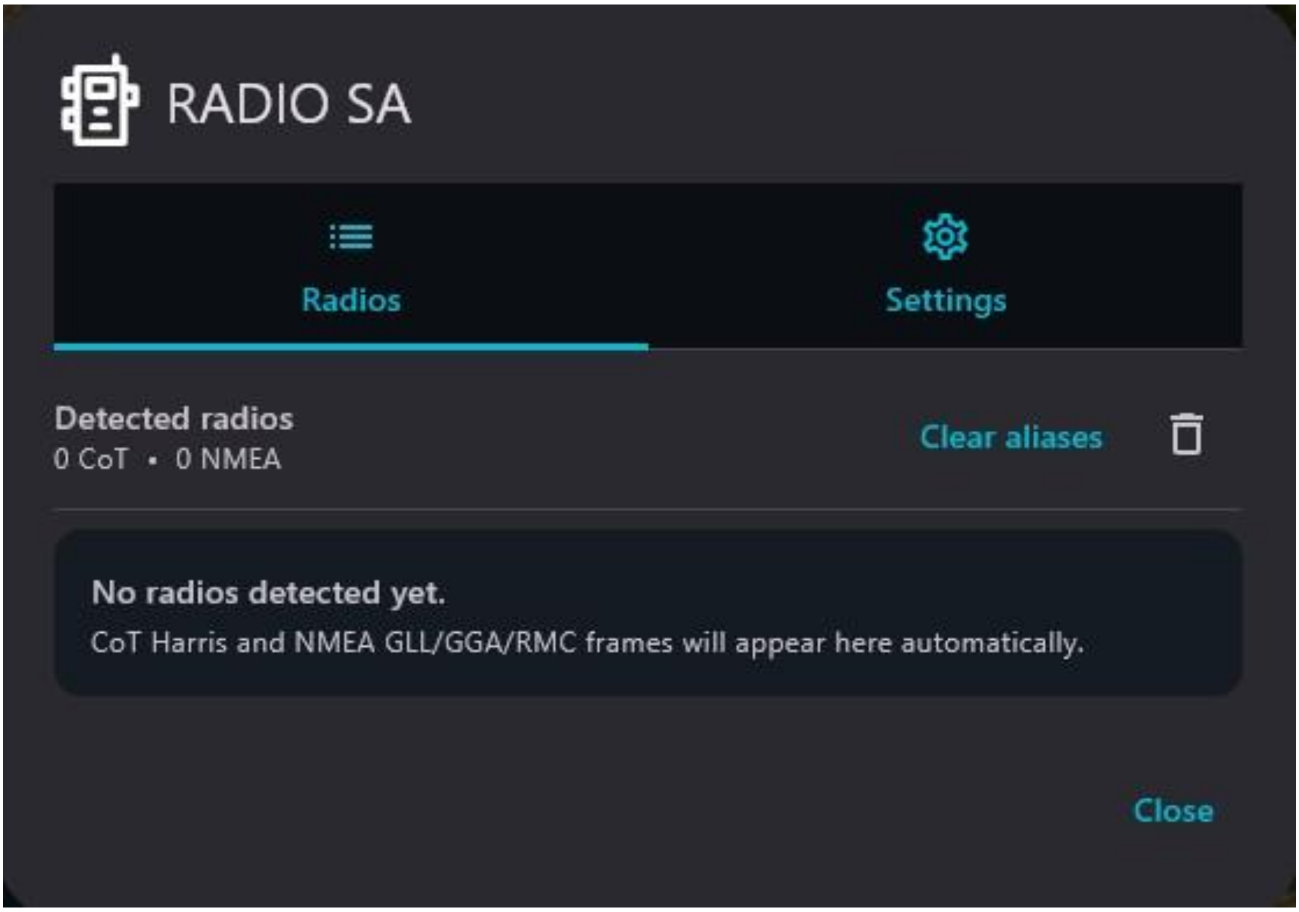

Figure 43 — List of detected radios and alias management.

Operating procedure

138. Wait for a supported radio message to be received.
139. Select the detected source and assign the correct callsign.
140. Verify that the track appears on the COP.
141. Use Clear aliases only when the associations must be rebuilt.

8. Camera, video and STANAG 4609 streams

Camera manages images and live transmissions, while Video configures received streams and their display on the COP.

8.1 Camera: Image on map

The Image on map tab allows an image file to be selected, an MGRS position and bearing to be set, and then a marker to be created on the map.

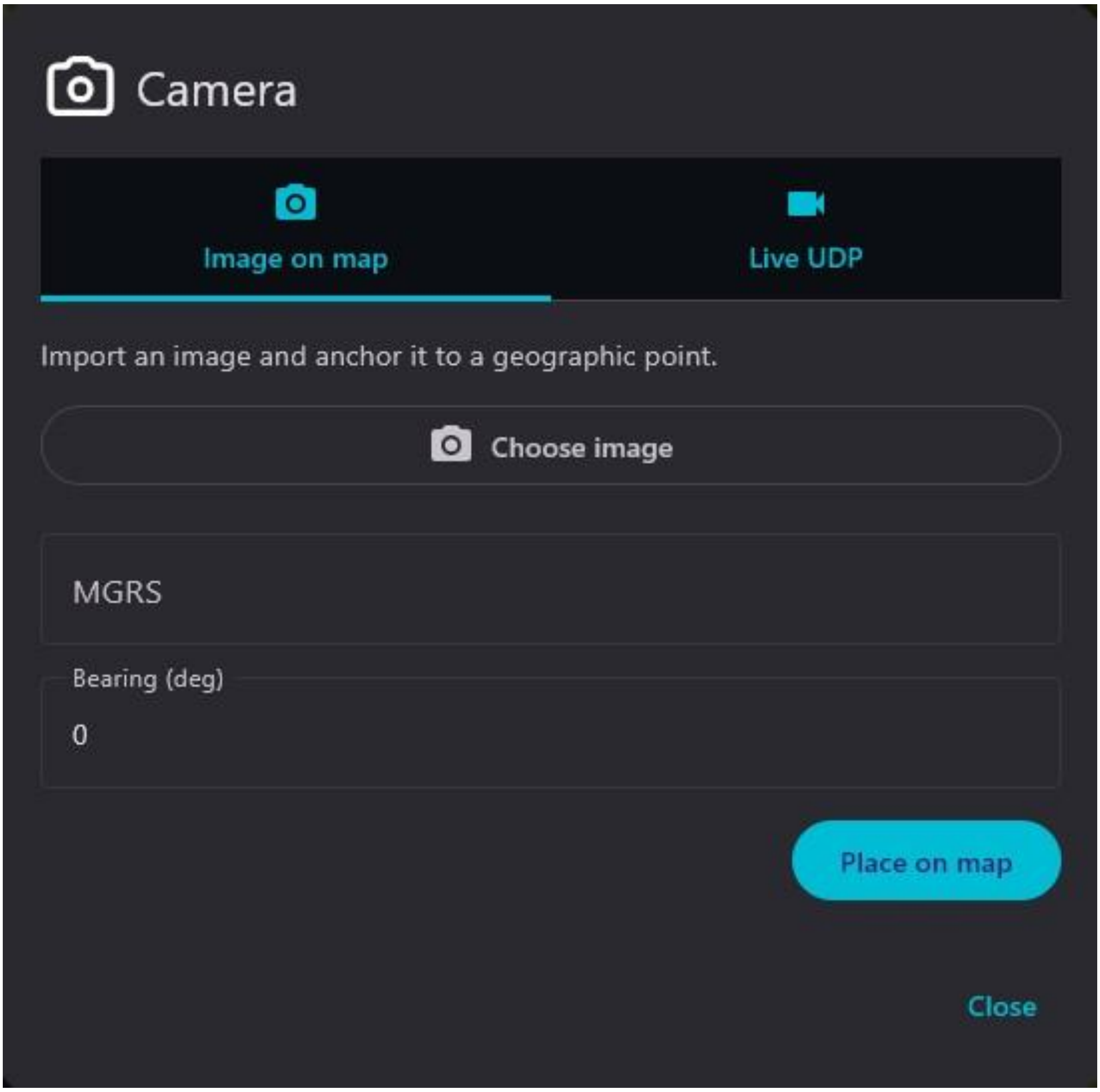

Figure 44 — Importing an image and placing it at a geographic point.

Operating procedure

142. Press Choose image and select the authorized file.
143. Verify or enter the MGRS position.
144. Set the bearing when required.
145. Press Place on map and check the marker.

OPERATIONAL NOTE The position associated with the image must be verified before sharing.

8.2 Camera: Live UDP

Live UDP transmits the internal or connected camera to a unicast or multicast destination. Camera selection, quality, address, port, interface and bitrate are available.

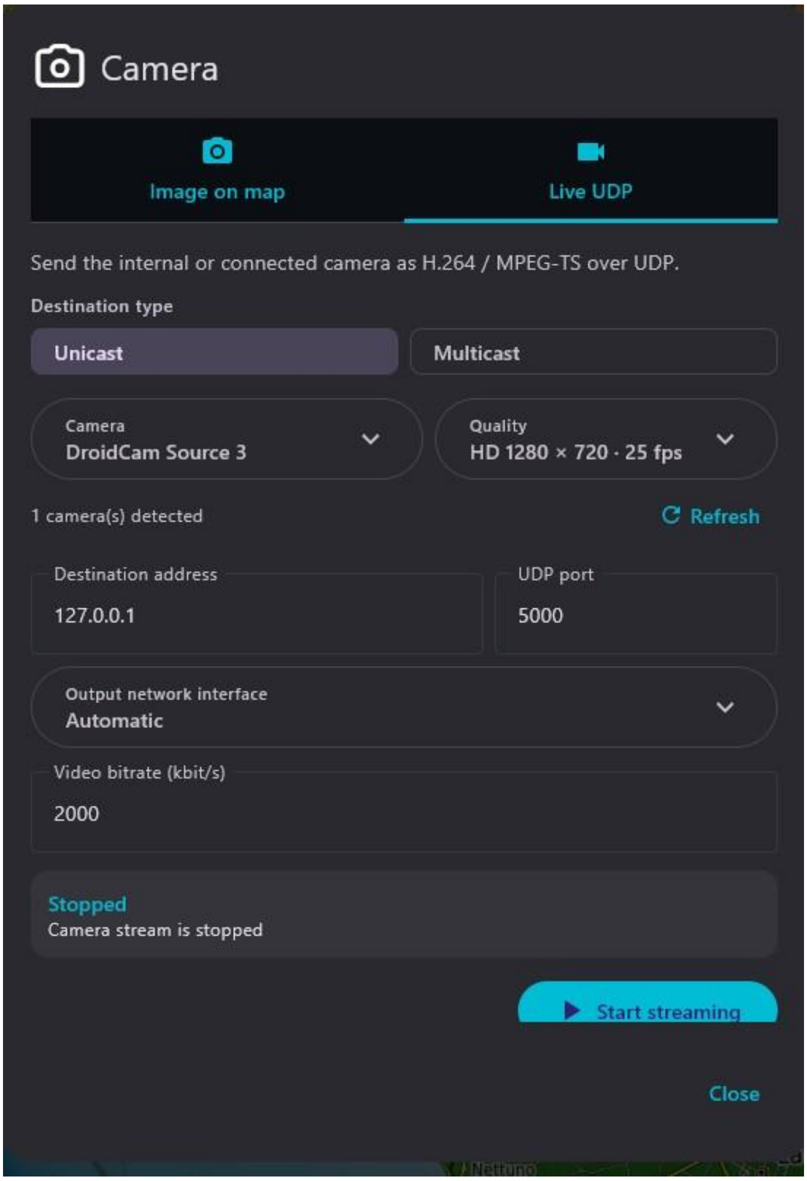

Figure 45 — H.264/MPEG-TS transmission over UDP with quality, destination and bitrate.

Operating procedure

146. Select Unicast or Multicast.
147. Choose the camera and quality.
148. Enter the Destination address and UDP port.
149. Select the output interface and bitrate.
150. Press Start streaming and verify reception.

CAUTION Reduce quality and bitrate on constrained links; stop the stream when it is no longer required.

8.3 Video stream list

Video shows the configured streams and the commands to start, edit or delete each entry. The note indicates that the KLV track can be managed separately from the video.

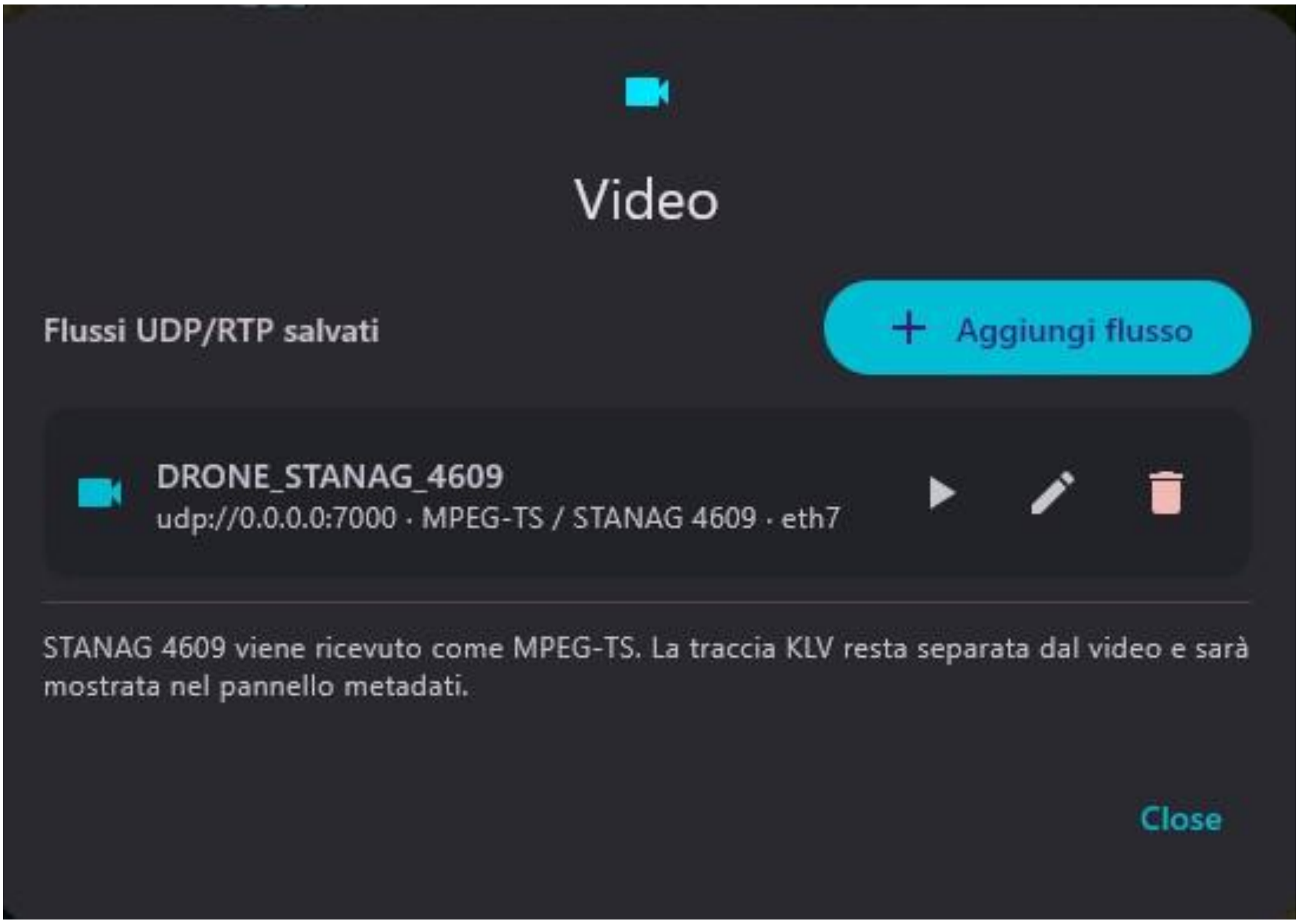

Figure 46 — List of saved UDP/RTP streams with start, edit and delete commands.

Operating procedure

151. Select the correct stream by name and URI.
152. Press Play to start reception.
153. Use Edit to correct the parameters.
154. Delete the stream only when it is no longer to be used.

8.4 Adding a stream

The configuration panel allows the alias, transport, format/payload, multicast group or unicast reception, port, interface, buffer, timeout and KLV handling to be set.

Figure 47 — Configuration of a UDP/RTP video stream with format, port, interface and buffer.

Operating procedure

155. Assign a clear name to the stream.
156. Select the correct transport and format.
157. Enter the multicast group or leave the field blank for unicast, as indicated by the screen.
158. Set the port, interface, buffer and timeout.
159. Press Save and start a test.

OPERATIONAL NOTE Ignore embedded KLV must be enabled only when the metadata track is managed separately.

8.5 STANAG 4609 without KLV geometry

When video is received but geospatial metadata is unavailable or invalid, the video window may be displayed without a platform icon, line of sight or footprint.

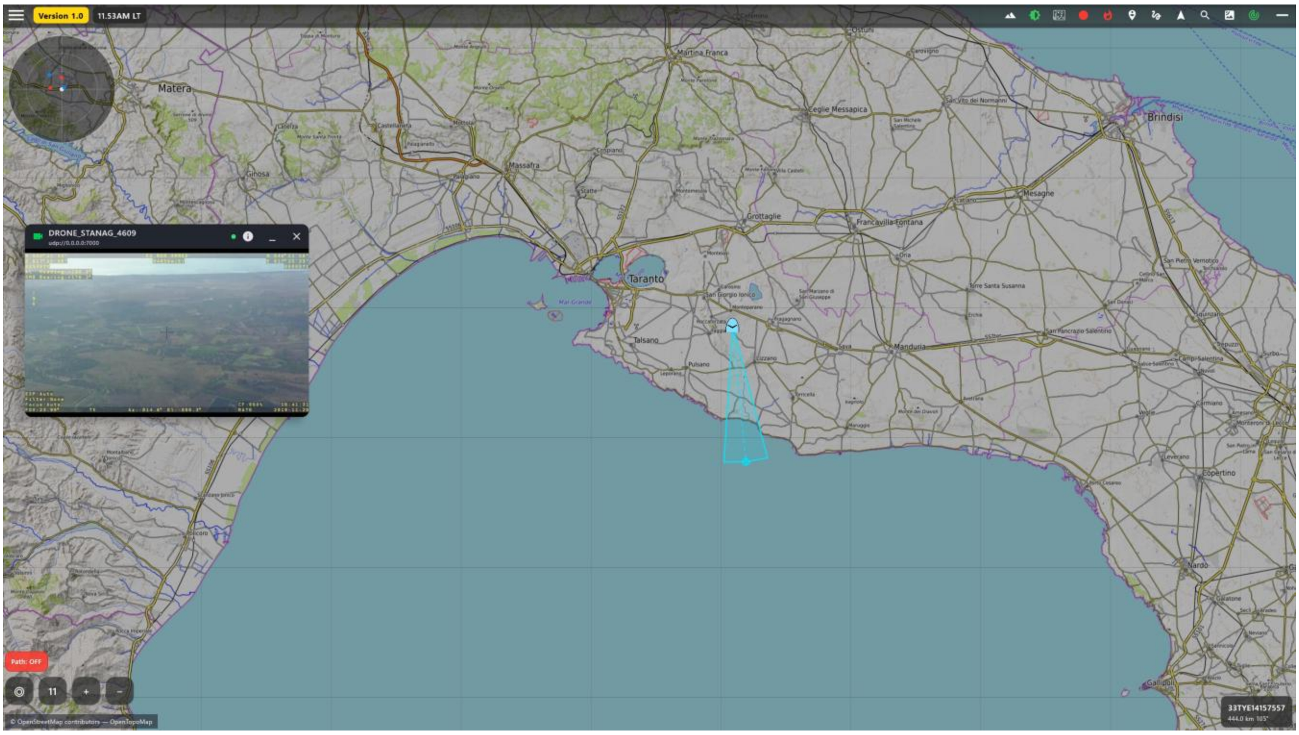

Figure 48 — STANAG 4609 stream displayed without full geospatial representation.

Operating procedure

160. Confirm that the video is stable.
161. Check for the presence of the KLV track.
162. Verify the timestamp and platform/sensor fields in the generator.

8.6 STANAG 4609 with KLV

With compatible KLV metadata, the map can show the platform position, line of sight and sensor footprint, as well as the decoded data panel.

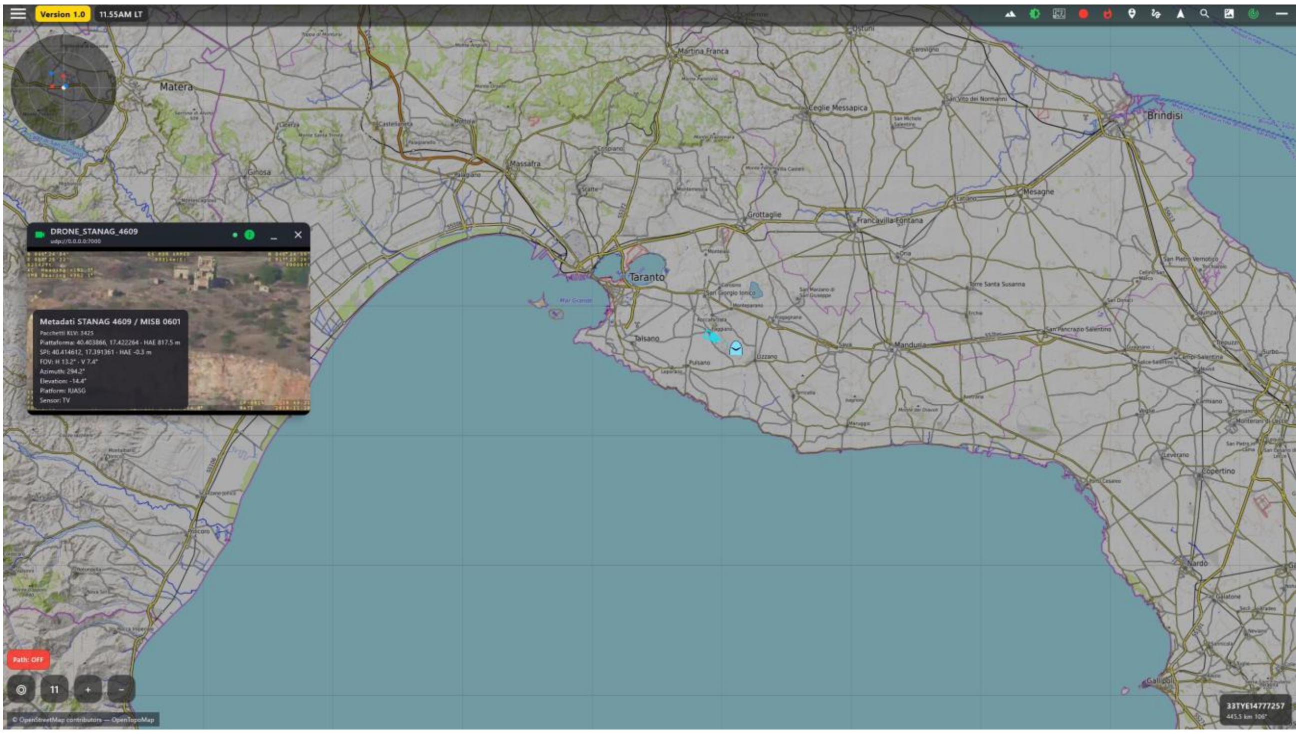

Figure 49 — STANAG 4609 stream with KLV metadata and sensor representation on the COP.

Operating procedure

163. Verify temporal consistency between the video and map.
164. Compare the platform position with a known source.
165. Check the orientation and footprint.
166. Stop operational use if the metadata is inconsistent.

CAUTION The map display depends on the correctness of the KLV fields and does not constitute an independent measurement.

9. Navigation and Drawing Tools

Navigation and drawing functions allow rapid references, measurements, fans, rings and operational graphics to be created on the COP.

9.1 Quick Navigation

Quick Navigation creates a line to a point or entity and displays azimuth and distance without building a complex route.

Figure 50 — Graphical link to a point with direction and distance.

Operating procedure

167. Enable Quick Navigation.
168. Select the destination point or entity.
169. Read the direction and distance along the line.
170. Disable the function when finished.

9.2 Range fan configuration

From the My position panel, an oriented fan can be defined by specifying azimuth, width and concentric distances.

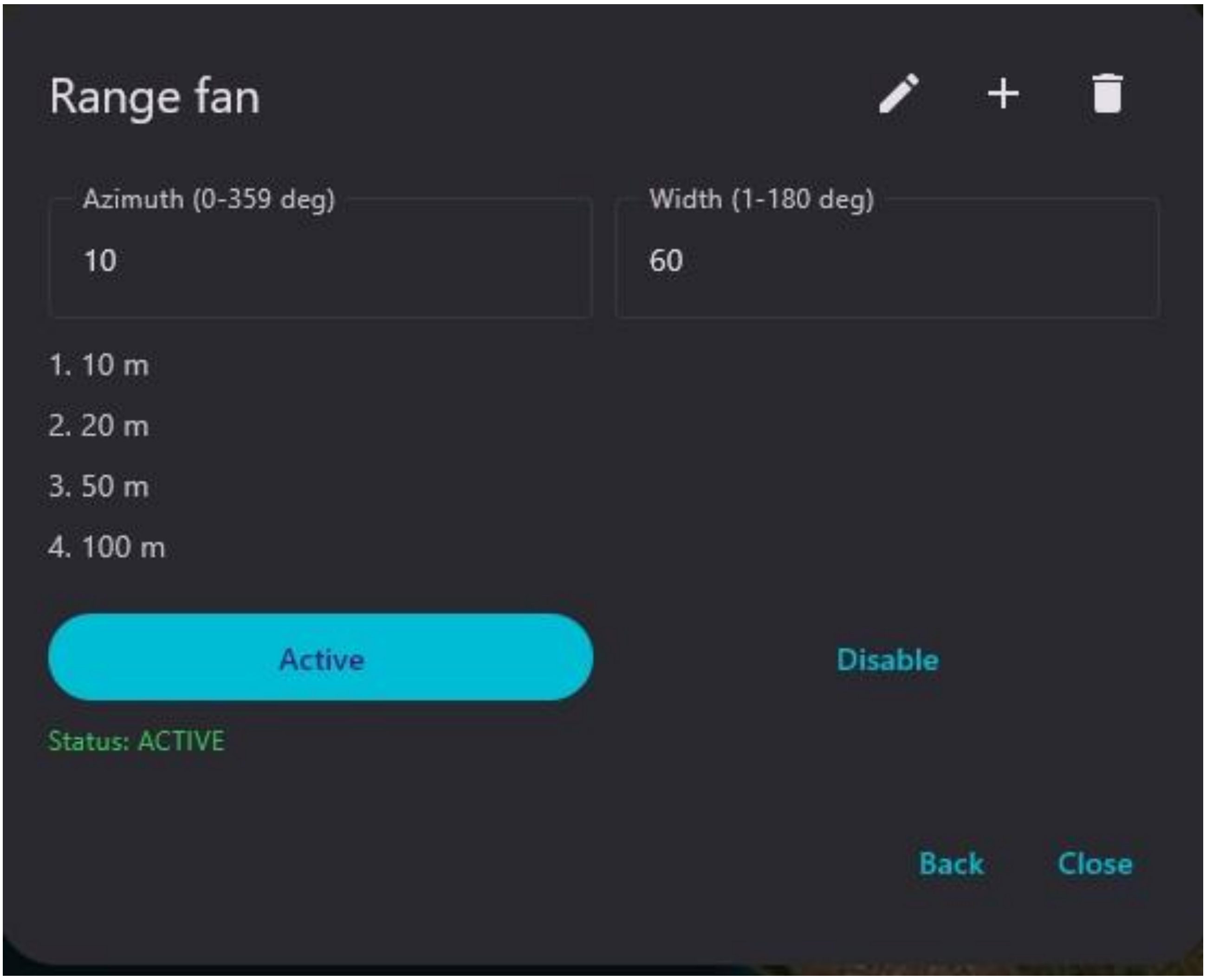

Figure 51 — Range fan azimuth, width and distance parameters.

Operating procedure

171. Set the azimuth in degrees.
172. Define the sector width.
173. Add or edit the ring distances.
174. Press Active and verify the result on the map.

9.3 Range fan on the map

The fan displays an oriented sector and concentric rings associated with the local position or a defined point.

Figure 52 — Range fan sector with distance rings.

Operating procedure

175. Check the origin, orientation and width.
176. Verify the distance labels.
177. Disable the fan when it is no longer valid.

CAUTION Range fan is a graphical representation and does not replace certified coverage, ballistic or visibility data.

9.4 Weapon ranges configuration

Weapon ranges allows multiple concentric distances to be configured, enabled and edited from the My position panel.

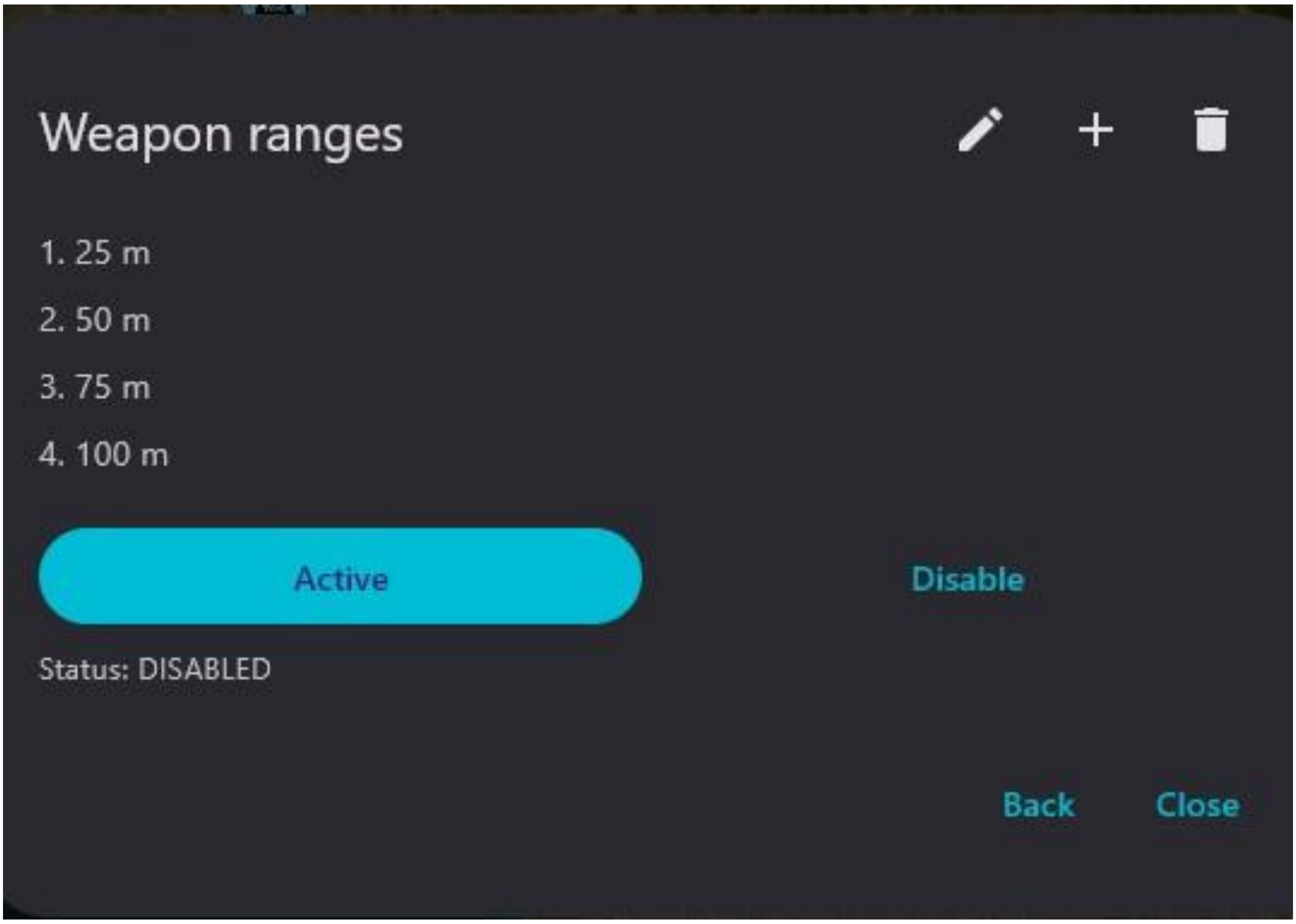

Figure 53 — Management of Weapon ranges rings from the local position.

Operating procedure

178. Enter the distances specified by the authorized configuration.
179. Press Active.
180. Verify the rings and labels on the COP.
181. Use Disable when the representation is not required.

9.5 Weapon ranges on the map

The rings provide an immediate visual reference around the vehicle. The displayed distances must be configured and interpreted according to the actual system represented.

Figure 54 — Concentric Weapon ranges rings displayed on the COP.

Operating procedure

182. Check that the center matches the correct position.
183. Read the labels before using the reference.
184. Remove or update the rings when the configuration changes.

CAUTION The graphic does not automatically calculate performance, obstacles, ballistics or environmental conditions.

9.6 Drawing Tools menu

Drawing Tools groups measurement and annotation functions. Each card shows the status or number of saved items.

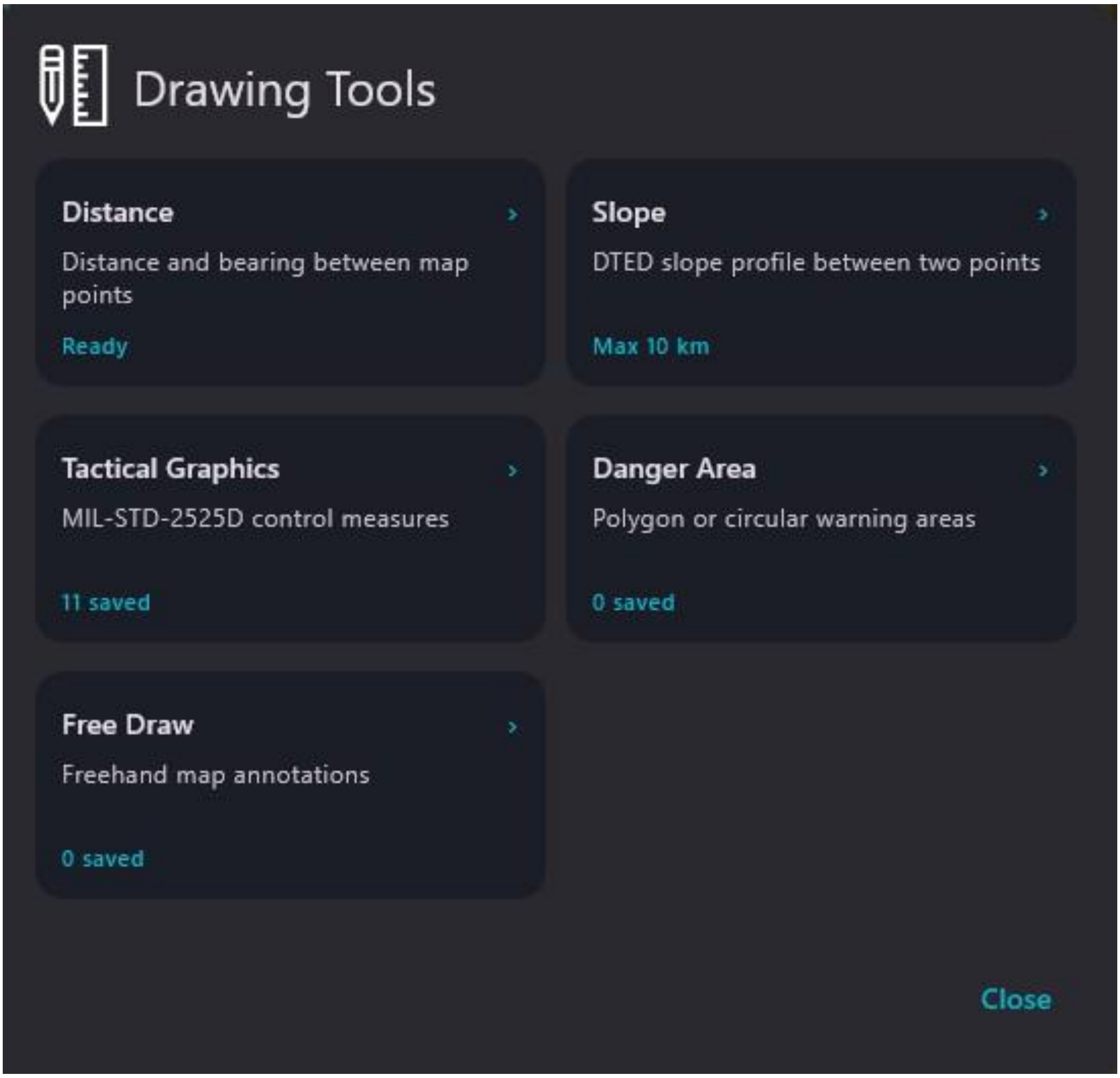

Figure 55 — Drawing Tools menu with Distance, Slope, Tactical Graphics, Danger Area and Free Draw.

Item	Purpose	Operating guidance
Distance	Measures distance and direction between points.	Use clearly identifiable points.
Slope	Calculates the elevation profile between two points.	Requires available DTED data.
Tactical Graphics	Creates control graphics.	Verify the affiliation and type.
Danger Area	Creates circular or polygonal warning areas.	Assign a clear name and validity period.
Free Draw	Adds freehand annotations.	Use simple, readable strokes.

9.7 Distance

Distance records the starting point and subsequent waypoints, displaying the total distance and values along the line.

Figure 56 — Linear or segmented measurement with total distance.

Operating procedure

185. Open Distance and select START.
186. Add the required waypoints.
187. Read the total distance.
188. Use Undo for the last point or Clear to delete the measurement.

9.8 Slope

Slope uses two points to show distance, elevations, elevation difference, gradient and the profile graph.

Figure 57 — Slope profile calculated using DTED data.

Operating procedure

189. Verify that the area is covered by DTED.
190. Select point A and point B within the indicated limit.
191. Read the elevations, difference and profile.
192. Press Clear for a new analysis.

OPERATIONAL NOTE Buildings, vegetation and local obstacles may not be represented by the elevation dataset.

9.9 Tactical Graphics

Tactical Graphics allows the catalog to be filtered by affiliation, the type to be searched and the selected graphic to be created on the map.

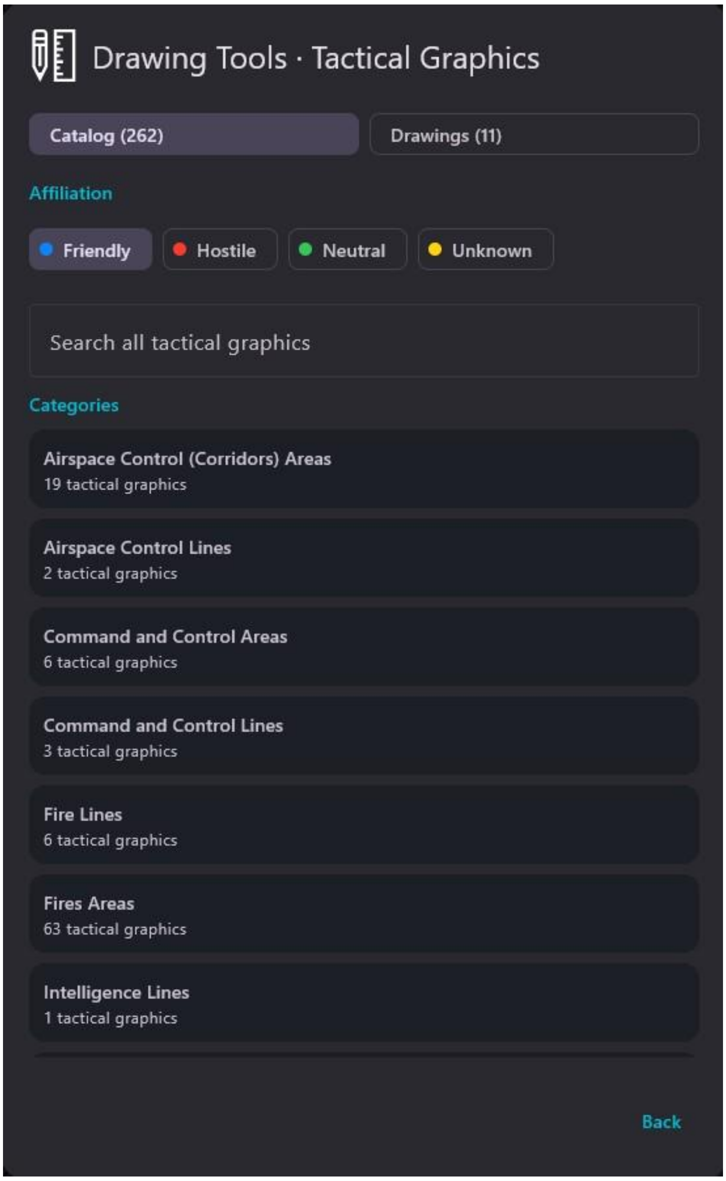

Figure 58 — Tactical Graphics catalog with affiliations, search and categories.

Operating procedure

193. Select Friendly, Hostile, Neutral or Unknown.
194. Use search or open a category.
195. Select the graphic type.
196. Enter the required points and save.
197. Verify the object in the Drawings tab.

CAUTION Always check the affiliation before saving.

9.10 Danger Area and Free Draw

Danger Area creates circular or polygonal warning areas; Free Draw allows freehand annotations. The image file does not contain dedicated detail screens, but both functions are available from the Drawing Tools menu.

198. Select the required shape or tool.
199. Enter the center/radius, vertices or freehand stroke.
200. Assign a clear designation and save.
201. Verify persistence and sharing according to the network configuration.
202. Remove objects that are no longer valid to avoid cluttering the COP.

10. Advanced Tools: Patrol, Path Drawing and LOS

Tools groups advanced functions for patrol planning, local path recording and visibility/propagation analysis.

10.1 Tools menu

The menu shows the three available functions and the summary status of each: saved patrols, Path Drawing enabled or disabled, and offline availability of the LOS module.

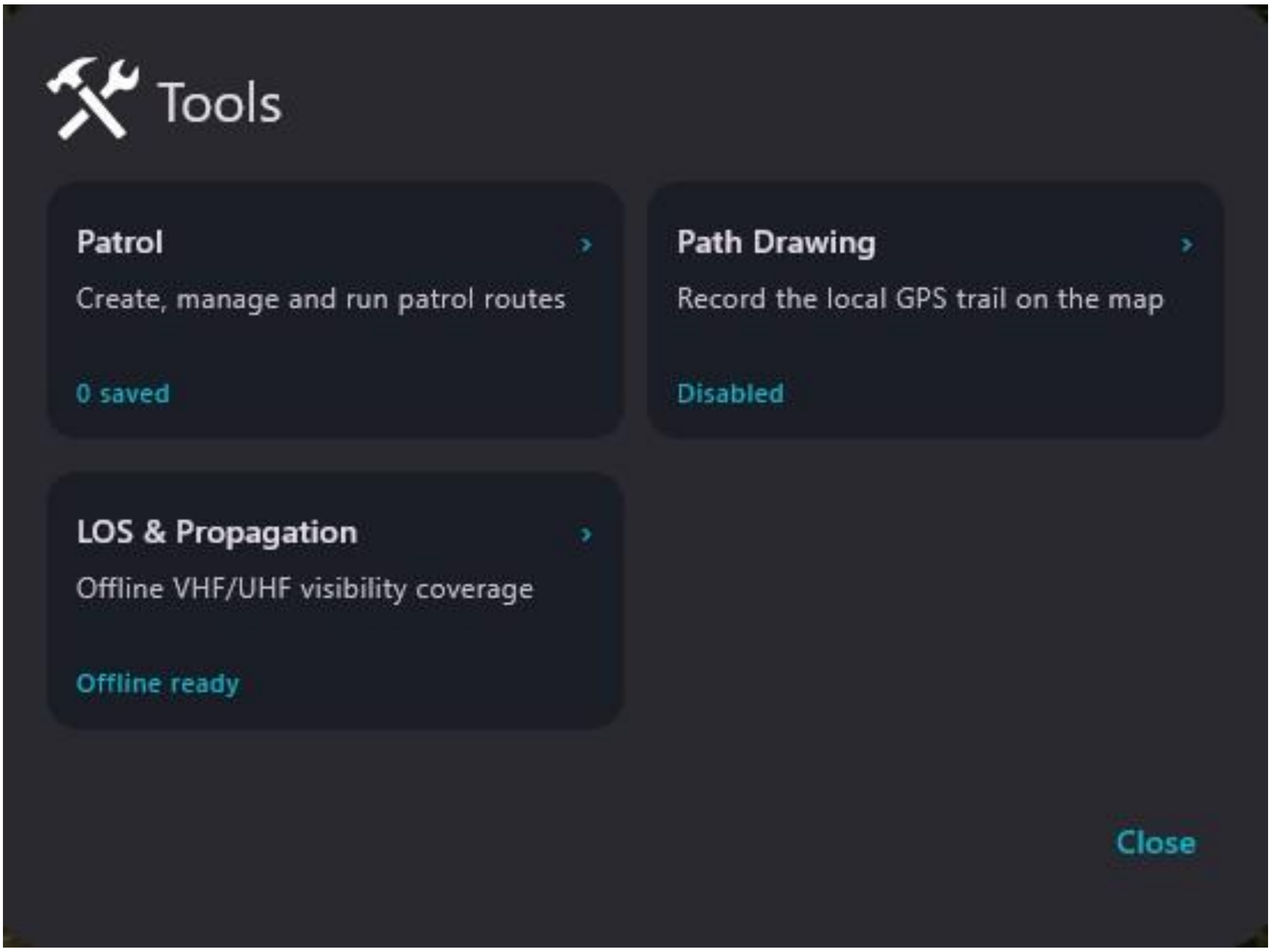

Figure 59 — Tools menu with Patrol, Path Drawing and LOS & Propagation.

Operating procedure

203. Open the required function.
204. Check the summary status before configuring.
205. Close the panel when the activity is complete.

10.2 Patrol management

Patrol allows a new patrol to be created or a saved route to be selected. The list shows the name, number of points, color and run or delete commands.

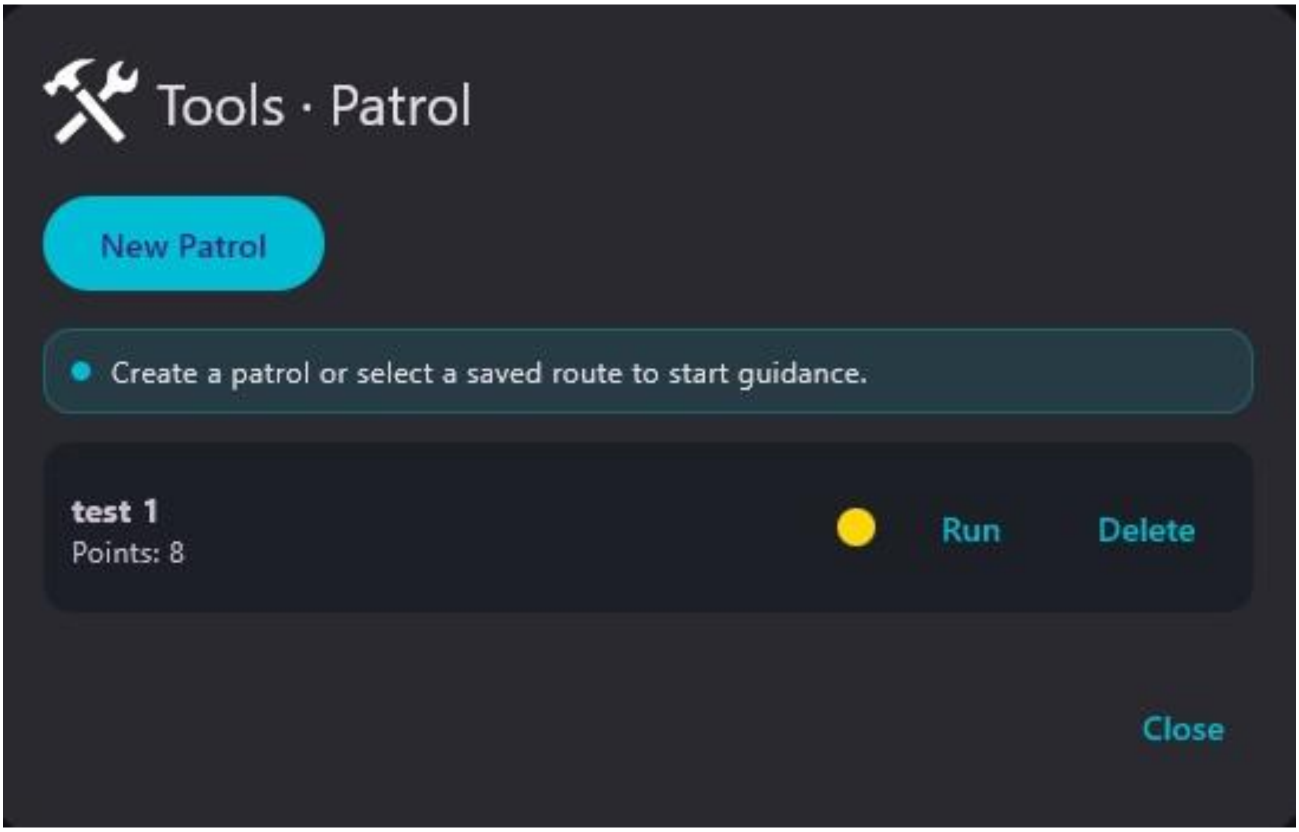

Figure 60 — List of saved patrols with Run and Delete commands.

Operating procedure

206. Press New Patrol to create a route.
207. Assign a name and enter the points on the map.
208. Save and verify the number of points.
209. Press Run to start guidance along the route.

10.3 Patrol on the map

During execution, the route is drawn on the map and the guidance panel shows the operational status.

Figure 61 — Patrol route displayed on the COP.

Operating procedure

210. Check that the route starts at the intended point.
211. Compare the waypoints with the actual situation.
212. Stop guidance if the route is no longer safe or valid.

CAUTION The map route does not replace assessment of the terrain, road network and actual conditions.

10.4 Path Drawing

Path Drawing records the vehicle GPS trail on the map. An enable switch, line width, color and Clear command are available.

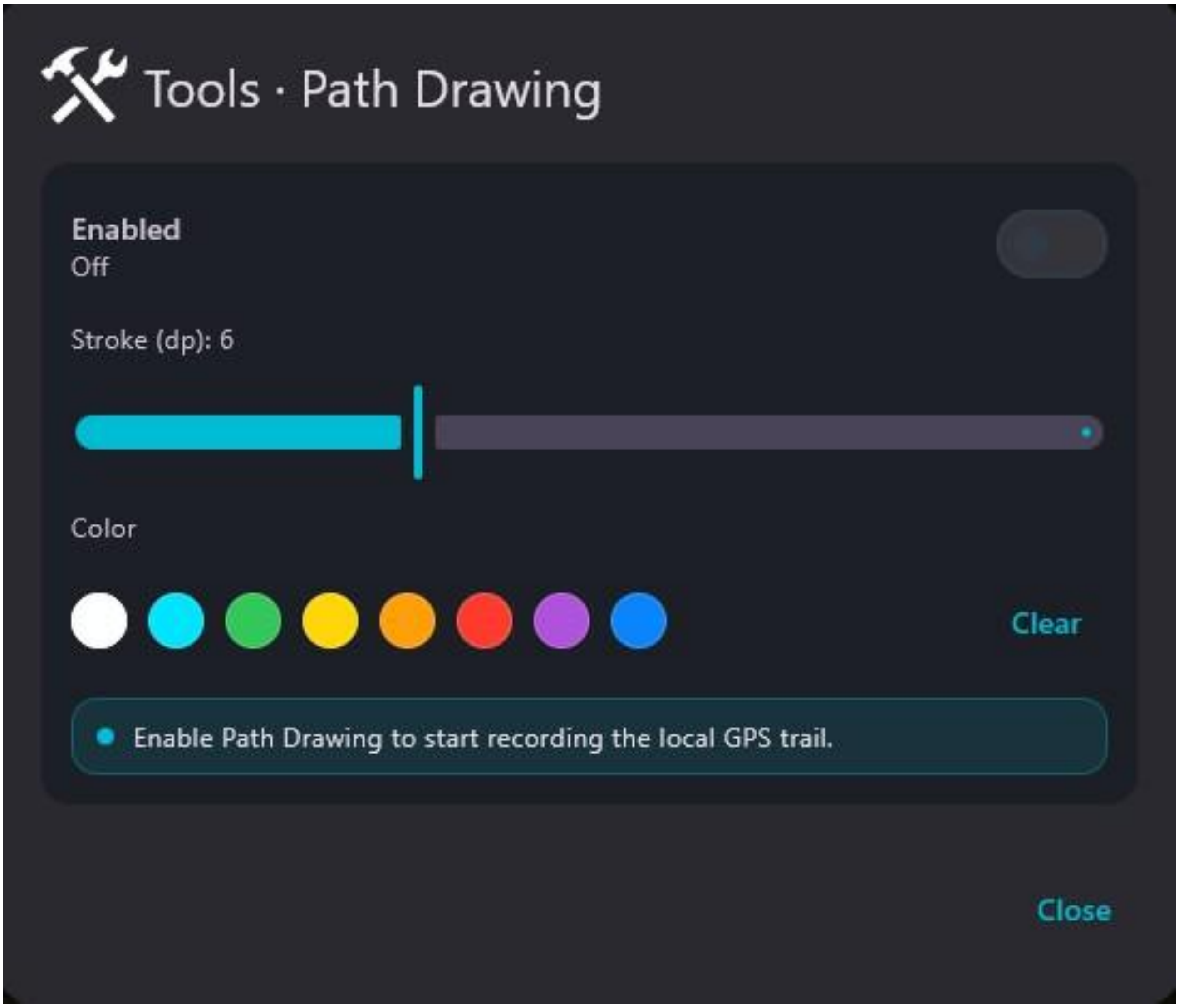

Figure 62 — Configuration of graphical recording of the local GPS path.

Operating procedure

213. Set a readable width and color.
214. Enable Path Drawing.
215. Verify that the path is recorded during movement.
216. Press Clear only when the trail is no longer required.

10.5 LOS & Propagation: parameters

LOS & Propagation allows the analysis center to be selected and the observer height, target height, transmitted power, frequency and environment type to be set. Leaving power and frequency blank performs line-of-sight analysis only.

Figure 63 — LOS/Propagation configuration with heights, power, frequency and environment.

Operating procedure

217. Set heights consistent with the observer and target.
218. Enter power and frequency only for propagation analysis.
219. Select the environment.
220. Press Pick center and select the point on the map.

CAUTION The result is an estimate based on the available data and does not replace certified radio planning.

10.6 LOS & Propagation: result

The colored area shows the estimate calculated around the selected point. The lower panel allows the display to be read or adjusted.

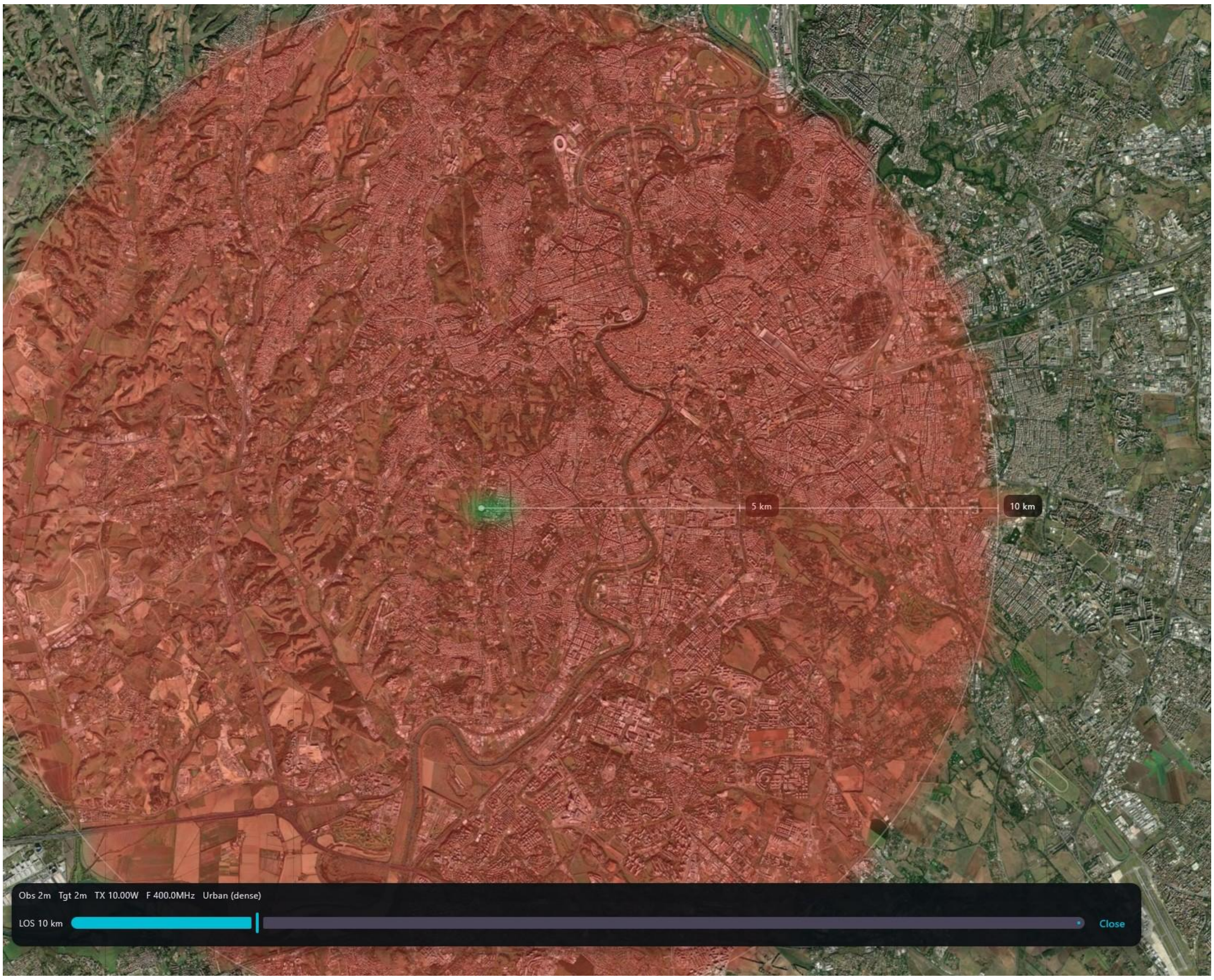

Figure 64 — Result of the LOS/Propagation analysis displayed on the map.

Operating procedure

221. Verify that the analysis center is correct.
222. Compare the coverage with known terrain and obstacles.
223. Repeat the analysis after changes in altitude, frequency or environment.
224. Close or remove the layer when it is no longer required.

11. Logistics management

Logistic represents the current status of resources defined by the administrator and allows operational values to be saved and transmitted.

11.1 Logistic module

The module allows an Excel configuration to be imported, the status to be sent and Current values to be updated. Max values come from the configuration and are read-only.

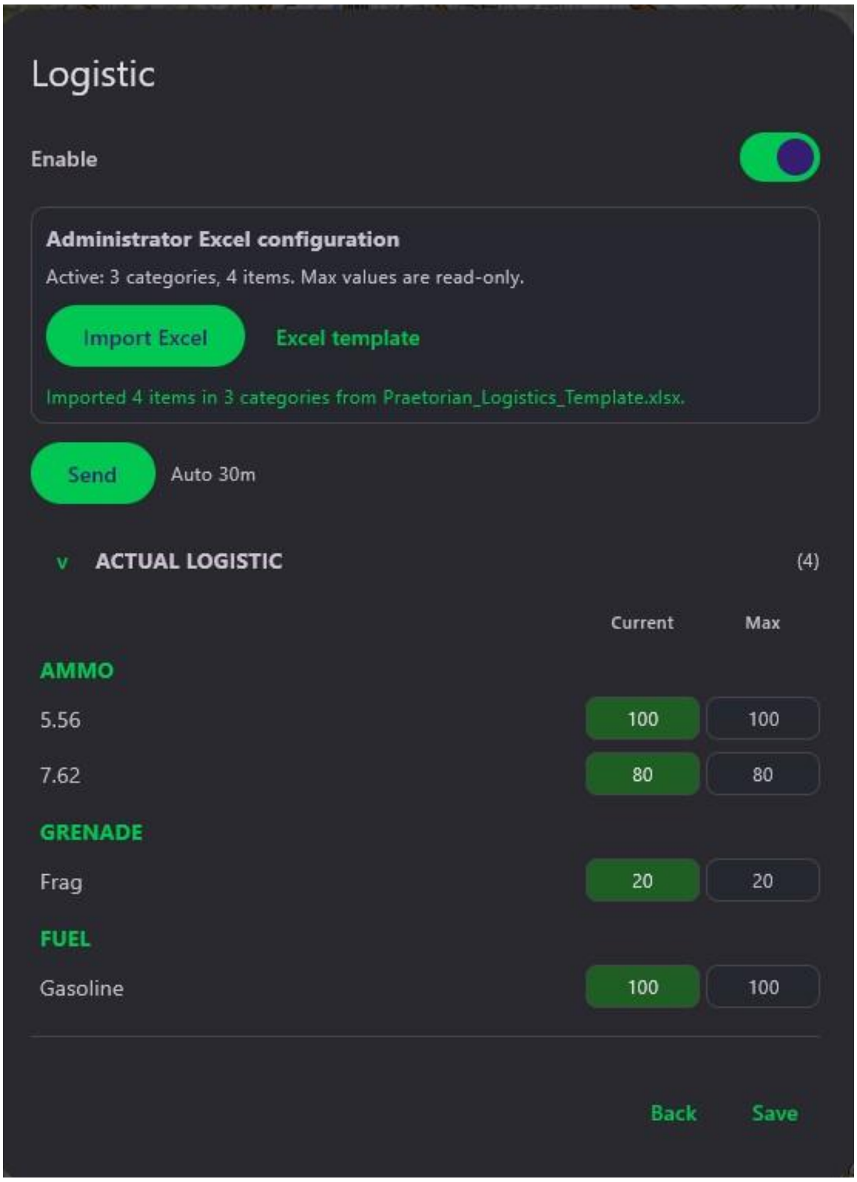

Figure 65 — Logistics configuration with Excel template, transmission and Current/Max values.

Operating procedure

225. Verify that the imported configuration is the intended one.
226. Update Current values after consumption or replenishment.
227. Check that no value exceeds Max.
228. Press Save.
229. Press Send when the status must be shared.

OPERATIONAL NOTE The quality of the logistics picture depends on timely operator updates.

12. Zeroize and data protection

Zeroize removes operational data stored locally. Complete deletion and selective deletion by category are available.

12.1 Panic Zeroize

Panic Zeroize deletes all Praetorian data locations and resets credentials. Selective Zeroize instead allows a single category to be selected.

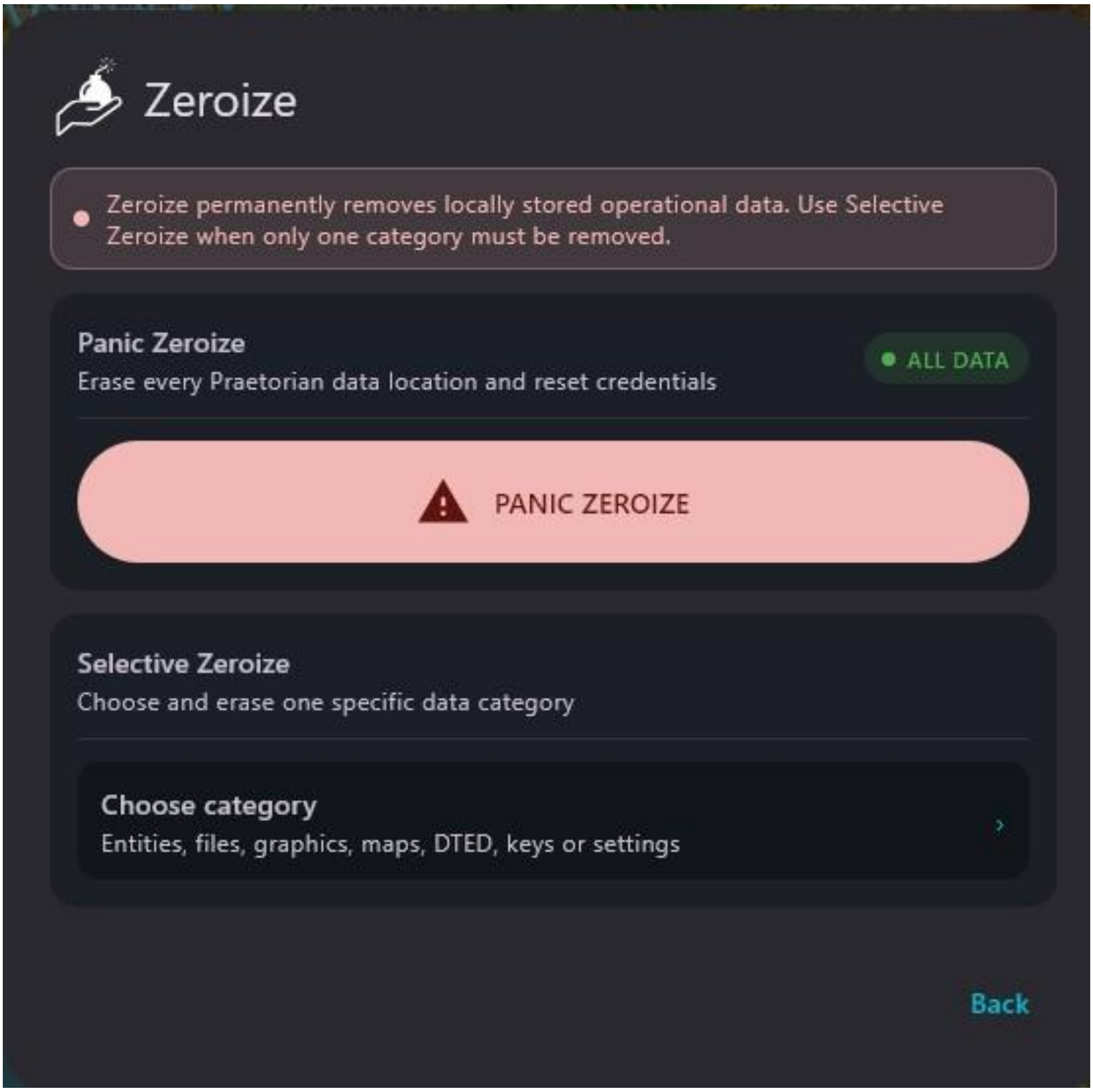

Figure 66 — Zeroize screen with Panic Zeroize and access to selective mode.

Operating procedure

230. Verify that the complete deletion order is authorized.
231. Open Zeroize and read the ALL DATA indication.
232. Press PANIC ZEROIZE.
233. Complete the confirmations only after the final check.
234. Consider the device non-operational until it has been reconfigured.

CAUTION Do not use Panic Zeroize to free space or resolve routine problems. The action is irreversible.

12.2 Selective Zeroize

Selective Zeroize shows the categories that can be deleted, including Remote entities, Local entities, Received files, Photo markers, Danger areas, Free draw and Tactical graphics.

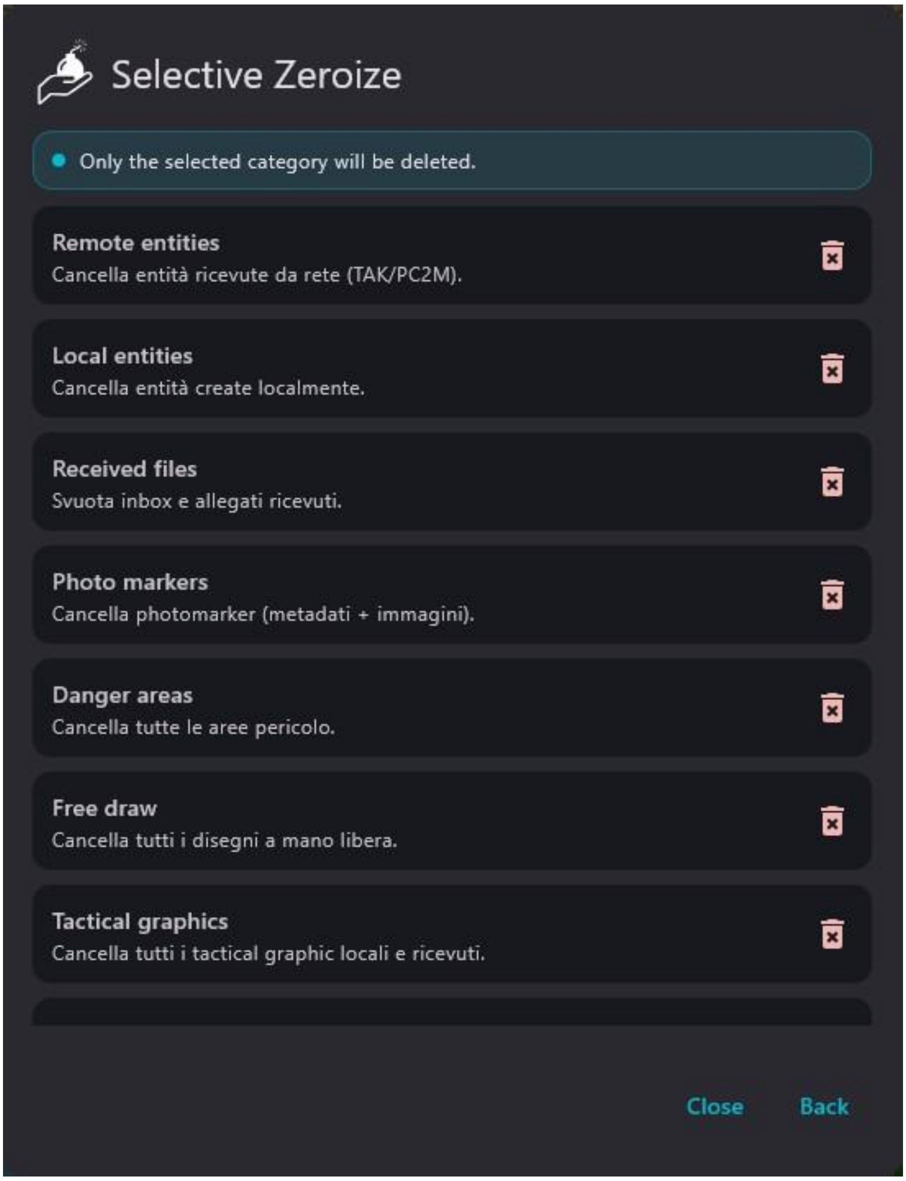

Figure 67 — Categories available for selective deletion.

Operating procedure

235. Select only one category.
236. Read the associated description.
237. Verify that the category is the authorized one.
238. Press the delete icon and confirm.
239. Reopen the module or the COP to verify the result.

CAUTION Selective deletion may also remove images, metadata or received objects associated with the category.

13. About, support and donations

The About and Donate screens provide information on the version, license, third-party components, manual, project support and donations.

13.1 About

About shows the application version and author, together with links to the license, third-party components, open sources and manual.

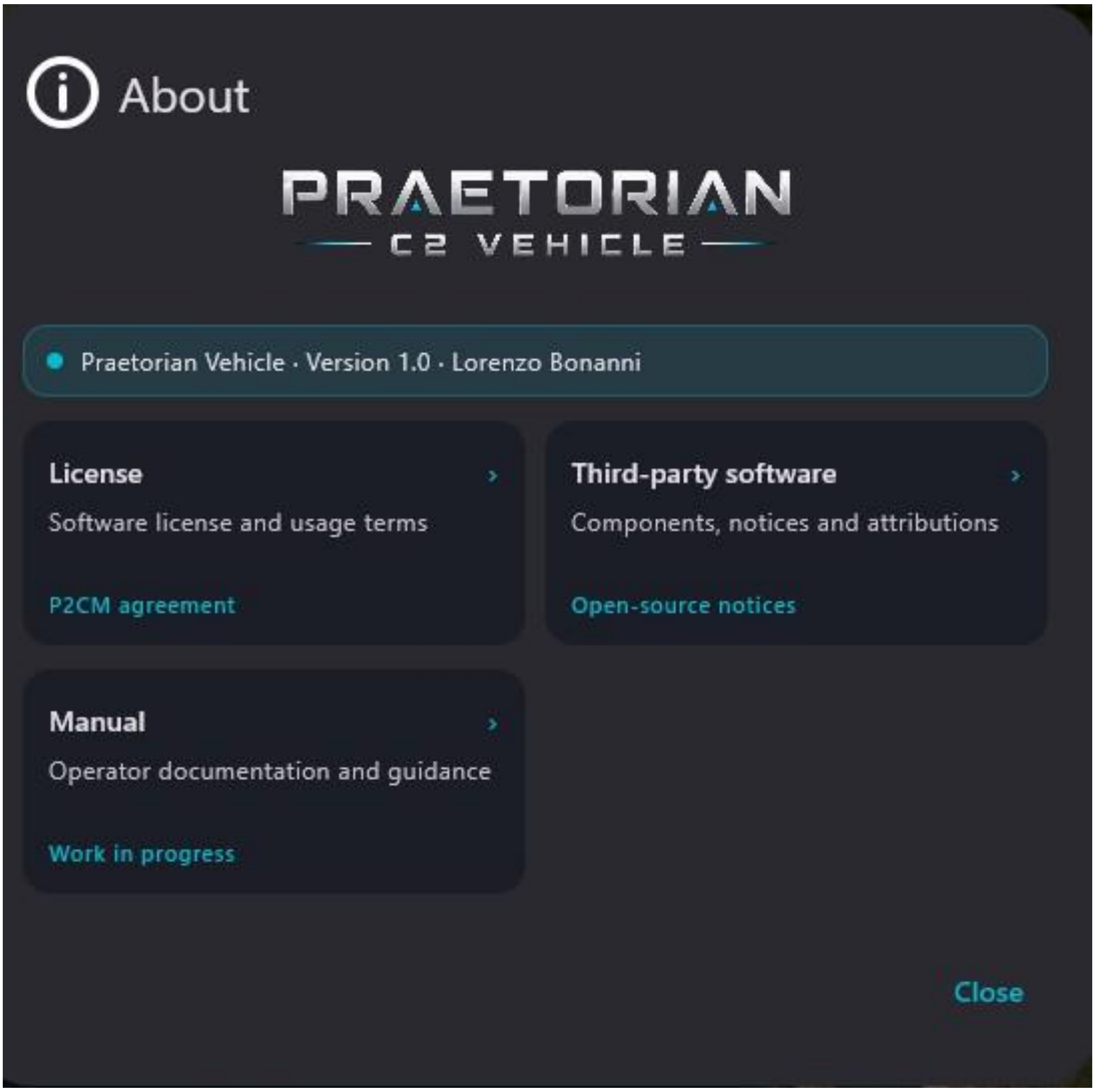

Figure 68 — About screen with version, license, third-party software and manual.

Operating procedure

240. Record the full version before requesting support.
241. Consult License and Third-party software for the applicable terms.
242. Open Manual for the available documentation.

13.2 Donate

Donate allows the development, maintenance and future improvements of the project to be supported through a QR code or PayPal link.

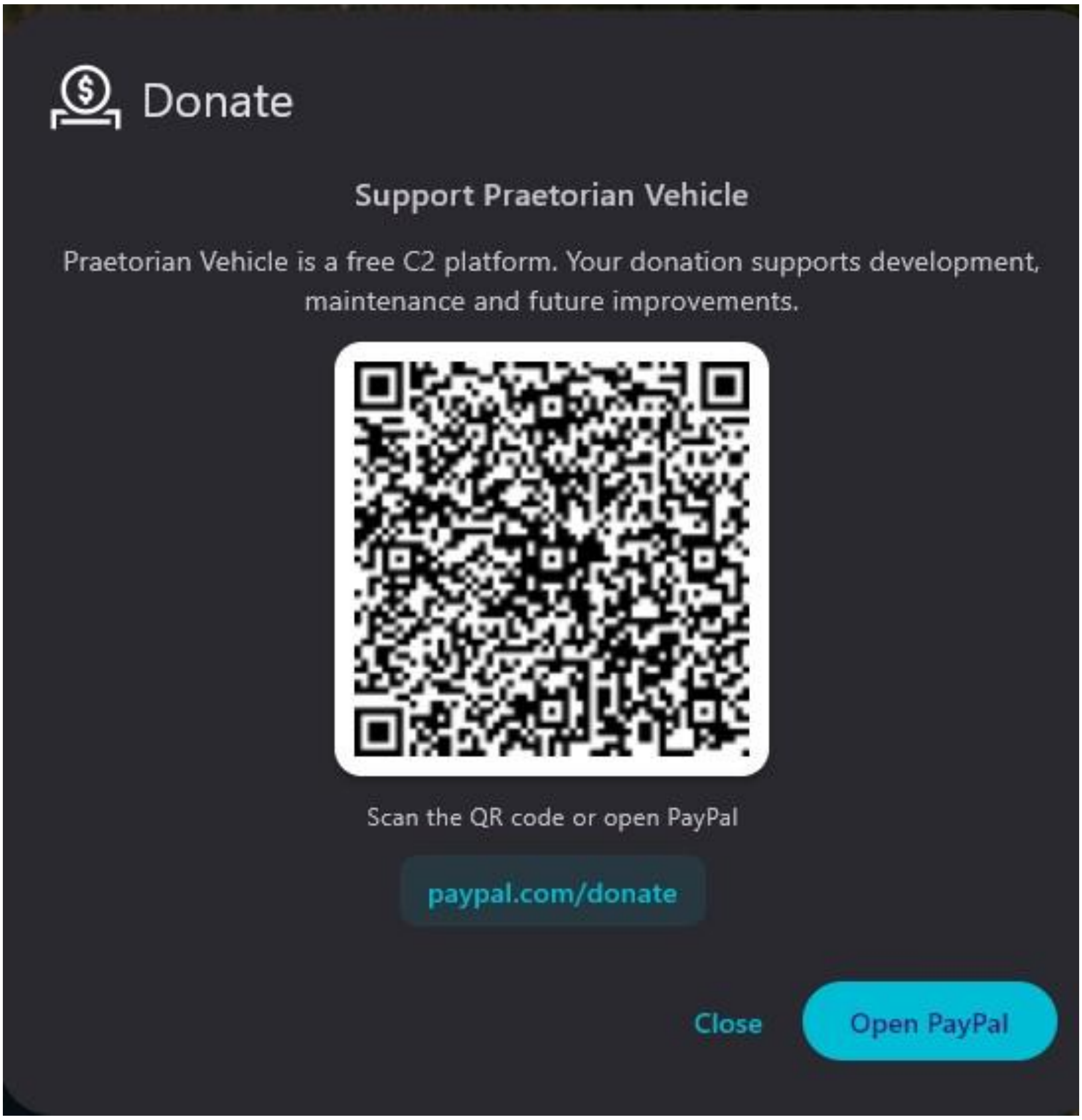

Figure 69 — Donate screen with QR code and PayPal access.

Operating procedure

243. Use the function only voluntarily and in accordance with organizational rules.
244. Verify that the opened link corresponds to the indicated page.
245. Close the panel to return to the COP.

14. Condensed operating procedures

14.1 Pre-mission checklist

- System started without errors; version and local time verified.
- Callsign, team color and local position correct.
- Map and elevation data for the area available.
- Network: interface, Input, Output and servers consistent with the architecture.
- LEGIO v2 configured and callsign assigned, when required.
- Expected tracks visible and traffic counters active.
- Alert presets, FORM MSG and contacts verified.
- Camera, video streams and addresses checked.
- Logistics configuration loaded and current values updated.
- Mission Data Packages imported and verified.

14.2 Sending a CONTACT/SITREP report

246. Confirm the event, position and time on the COP.
247. Open FORM MSG and select the required template.
248. Complete the mandatory fields.
249. Verify the coordinates, sender and recipient.
250. Use Save for a second check or Send to transmit.
251. Check the Sent tab.

14.3 Unicast video transmission

252. Confirm the receiver IP address and UDP port.
253. Open Camera > Live UDP and select Unicast.
254. Choose the camera, quality, interface and bitrate.
255. Start the stream and request confirmation of reception.
256. Stop transmission when finished.

14.4 Mission closeout

- Stop video streams and functions that are no longer required.
- Save or transmit the final authorized reports.
- Update the final logistics status.
- Remove temporary graphics and layers.
- Apply Selective Zeroize or Panic Zeroize only when ordered.
- Close the application and secure the device.

15. Troubleshooting

CONTROLLED DIAGNOSIS Record the initial status, change only one parameter at a time and verify the effect on the counters, COP and destination system.

Item	Purpose	Operating guidance
No tracks received	Received does not increase; COP unchanged.	Check My IP, Input, firewall, addresses/ports and source.
TAK Server offline	Server offline indicator.	Check the network, host, port, trust store and client certificate.
LEGIO not active	Invalid workbook or callsign.	Import the correct workbook, select the callsign and save.
Slow LEGIO message	Delayed delivery.	Compare with the profile latency; avoid retransmissions and prefer Text/COP delta.
RADIO SA not visible	No radio on the COP.	Enable RADIO SA, verify Input and assign an alias/callsign.
Camera not detected	Camera list empty.	Check the connection and permissions, then refresh the list.
Video not received	Active stream without an image.	Check transport, address/group, port, interface, firewall and format.
STANAG without drone icon	Video visible without geometry.	Check the KLV track, timestamp and platform/sensor fields.
Slope unavailable	Profile missing or error.	Check DTED, area coverage and the distance between the points.
Data Package not visible	Repository or list empty.	Press Query Server/Refresh and check the connection.
Incorrect logistics values	Current inconsistent or Max cannot be edited.	Correct Current; Max is derived from the Excel configuration.
Deletion performed by mistake	Local data missing.	Inform the responsible person and restore only from authorized sources.

16. Glossary and quick command reference

16.1 Essential glossary

Item	Purpose	Operating guidance
COP	Common Operational Picture.	Common operational picture integrating maps, tracks, graphics and mission information.
C2	Command and Control.	Functions and systems for command, control and coordination.
CoT	Cursor on Target.	Exchange format for tactical and geospatial events in the TAK ecosystem.
TAK	Team Awareness Kit ecosystem.	Family of interoperable applications and services for situational awareness.
SA	Situational Awareness.	Situational awareness derived from positions, events and reports.
LEGIO v2	Praetorian point-to-point narrowband channel.	Optimized for COP messages over narrowband, high-latency networks.
KLV	Key-Length-Value.	Metadata encoding used in STANAG video streams.
STANAG 4609	NATO standard for digital motion imagery.	May include KLV geospatial metadata.
DTED	Digital Terrain Elevation Data.	Digital elevation data used by Slope and terrain analysis.
MGRS	Military Grid Reference System.	Grid reference system used in operational messages.
Unicast	Transmission to a single address.	Used for a specific recipient.
Multicast	Transmission to an IP group.	Requires a consistent network and interface.
MTU	Maximum Transmission Unit.	Maximum packet size on the link.
Jitter	Variation in delivery delay.	May affect the regularity of messages and streams.

16.2 Quick command reference

Item	Purpose	Operating guidance
Close	Closes the current panel.	Does not necessarily cancel changes that have already been saved.
Back	Returns to the previous screen.	Allows the operator to remain in the main module.
Save	Stores the configuration or local content.	Verify the displayed status after the command.
Send	Transmits content or a message.	Check the recipient, network and data.
Refresh	Reloads devices, data or status.	Use it after connections or external changes.
Clear	Removes items from the current session.	May not delete objects that have already been saved.
Undo	Cancels the last point or step.	Available only in some tools.
Enable	Enables the module.	May start configured reception or transmission.
Import	Loads a configuration or package.	Verify the source and version.
Query Server	Queries the connected server.	Requires connectivity and authorization.
Start streaming	Starts UDP video.	Check bandwidth, address and port.
Panic Zeroize	Deletes all local data.	Irreversible action reserved for authorized procedures.

Appendix A — Data to record before requesting support

Item	Purpose	Operating guidance
Software version	Badge and About screen.	Report the complete value.
Time of the event	Date, local time and time zone.	Record the exact time of the anomaly.
Affected module	Open menu and submenu.	Indicate the operator sequence.
Network status	Server, counters, interfaces and links.	Mask sensitive data when required.

Item	Purpose	Operating guidance
Evidence	Authorized screenshot or log.	Exclude passwords, private keys and classified data.

Installers, manuals, data sheets, configurations and future integrations are published through the project website and GitHub releases. Always use the version specified by your organization and verify file integrity when official checksums are available.

Project website: moicanox.github.io/Praetorian-C2-Vehicle/

Support email: praetorian.c2.it@gmail.com